

# FinFETs

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# Course content

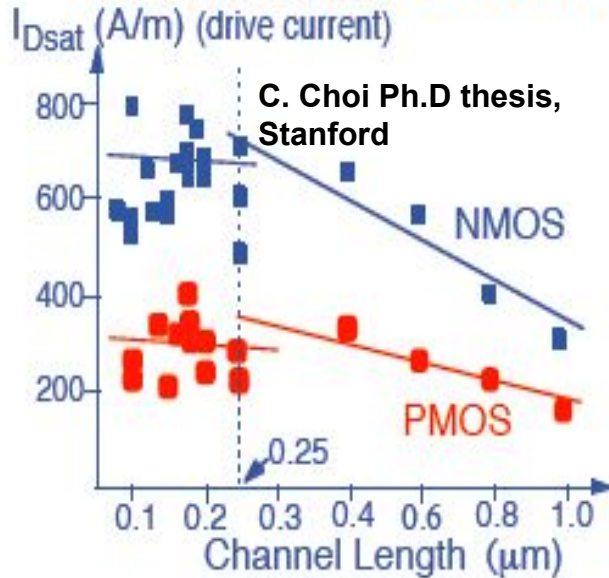
2

- Review of CMOS scaling. Problems with traditional (Dennard) scaling.
- Basic Quantum mechanics and Silicon band structure
- Mobility enhancement techniques (introduced in high volume manufacturing in 2003)
- High k gate dielectrics and metal gates (introduced in 2007)
- FinFETs (introduced in 2012)
- High mobility channel materials (introduced in 2011)
- Nanosheet transistors (introduced in 2024 Samsung Galaxy Watch7)
- Layout dependent effects (became a serious issue since late 2000s)

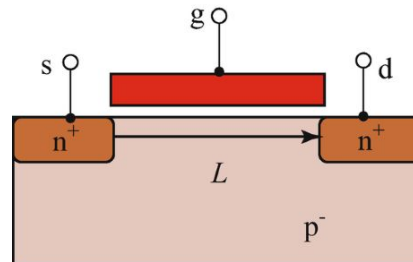
# Three horsemen of the apocalypse

3

## Constant OFF current Limit



Saturation of  $I_{dsat}$   
@ constant  $I_{off}$



IEEE ELECTRON DEVICE LETTERS, VOL. 18, NO. 5, MAY 1997

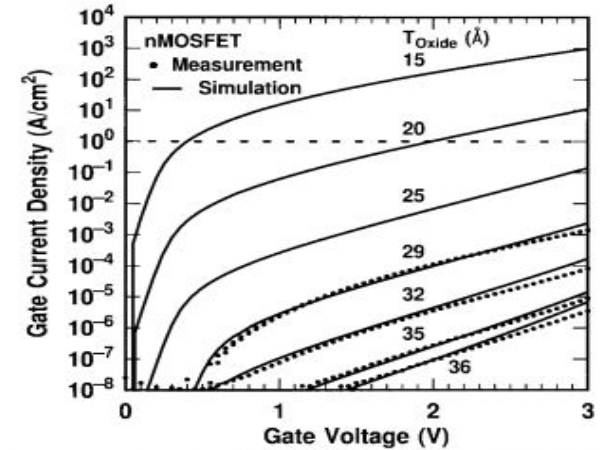
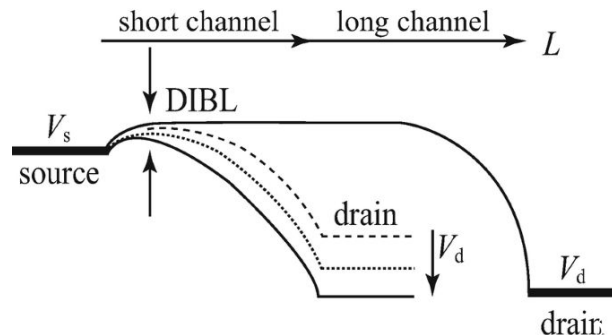


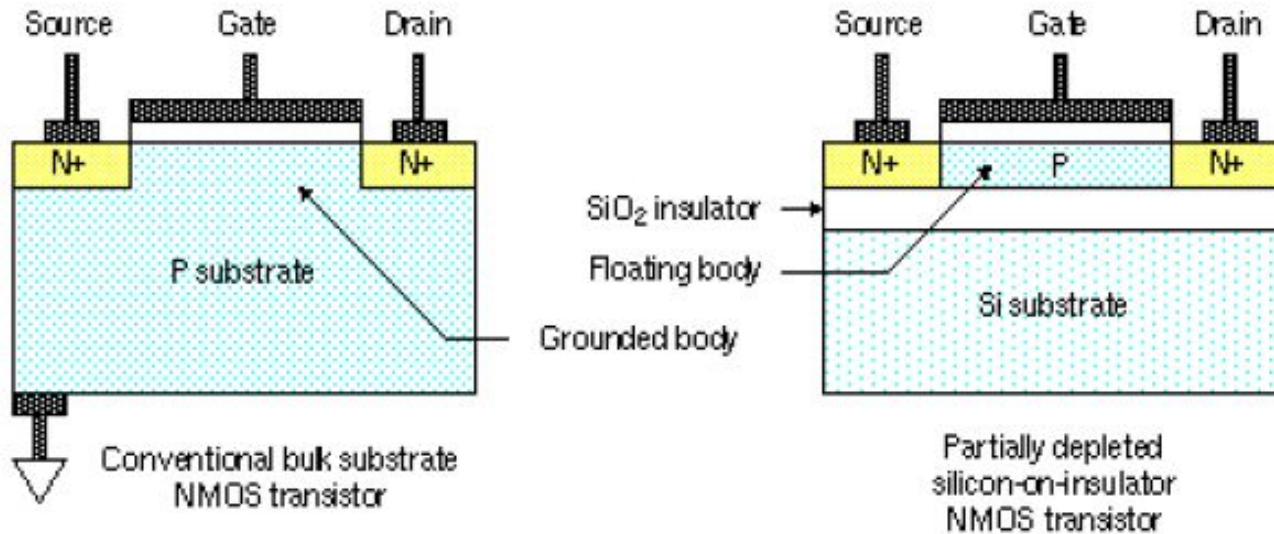
Fig. 2. Measured and simulated  $I_G-V_G$  characteristics under inversion conditions of four nMOSFET's. The dotted line indicates the 1 A/cm<sup>2</sup> limit for leakage current as discussed in the text.

Gate Leakage current  
due to tunneling



Worsened gate control

# Bulk vs SOI

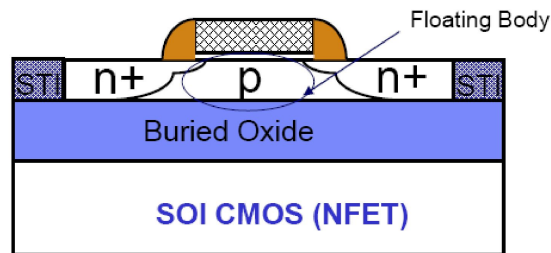


SOI could be partially depleted (PD) SOI or fully depleted (FD) SOI depending on whether the silicon body is fully depleted or not. PDSOI has been used in commercial production ( $t_{Si} > 70\text{-}100\text{nm}$ ) since 1990s. FDSOI has been used since 2010s

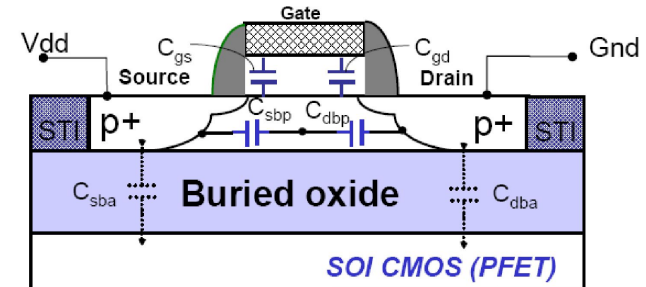
# SOI Value Adders

5

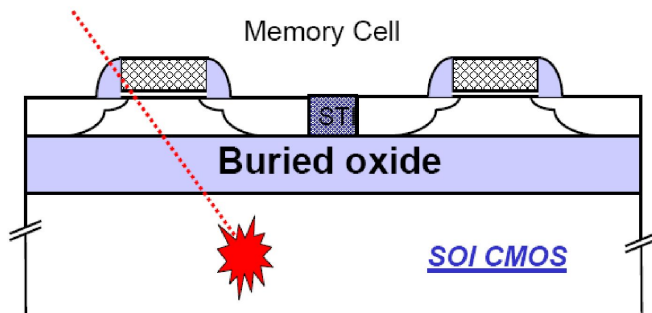
- Superior Isolation
  - No Well Taps required
  - Smaller circuit footprint
  - Latch-up free operation



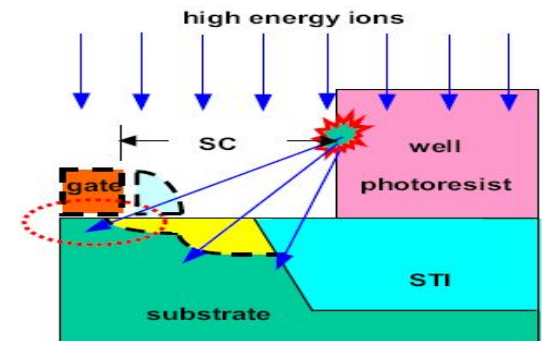
- High Speed Switching
  - 2x lower junction perimeter capacitance
  - Improved Short-Channel Effects



- Less susceptibility to soft errors
  - SER reduced by 5-7x
  - Low power high reliability

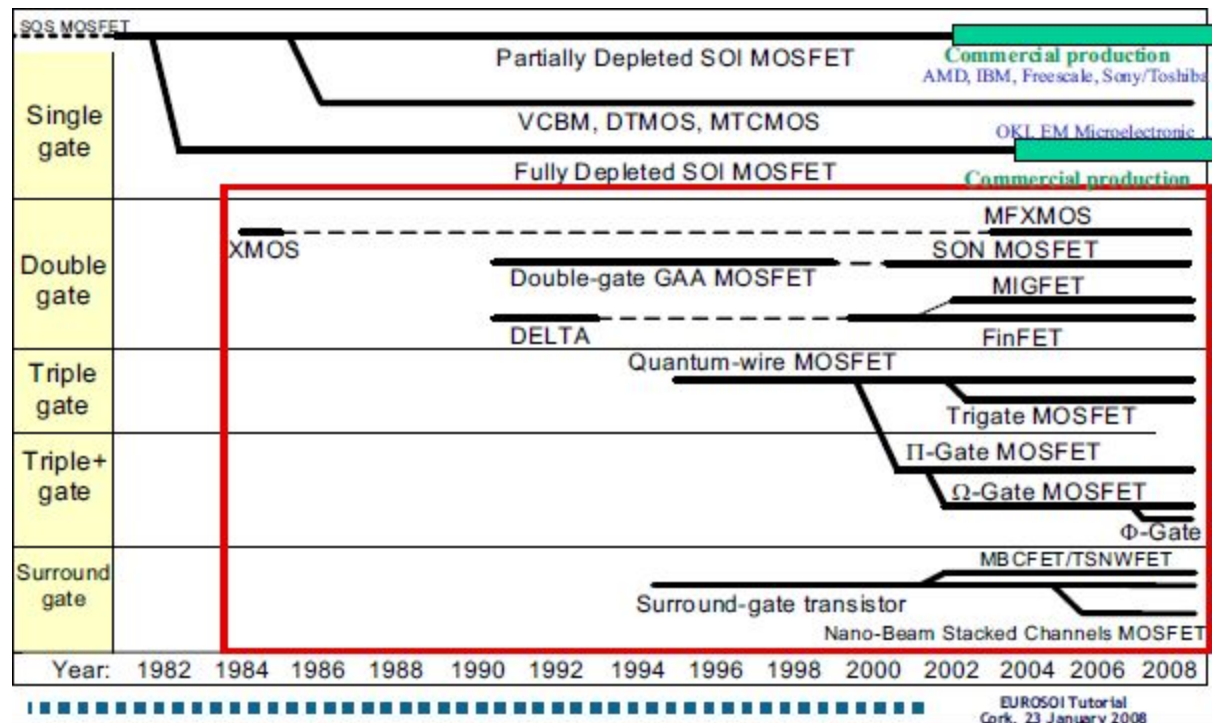


- Reduced Variation within Die
  - Short-channel  $V_t$  roll-off suppression
  - Less variation in  $V_t$  vs  $L_{gate}$ .
  - No Well-Proximity Effects (no deep wells in SOI)



# History

- T. Sekigawa and Y. Hayashi (1984) showed significant short channel improvement by adding addnl gate (XMOS)
- First realization was DELTA MOSFET by Hitachi (1989)



# Other configurations of MuGFETs

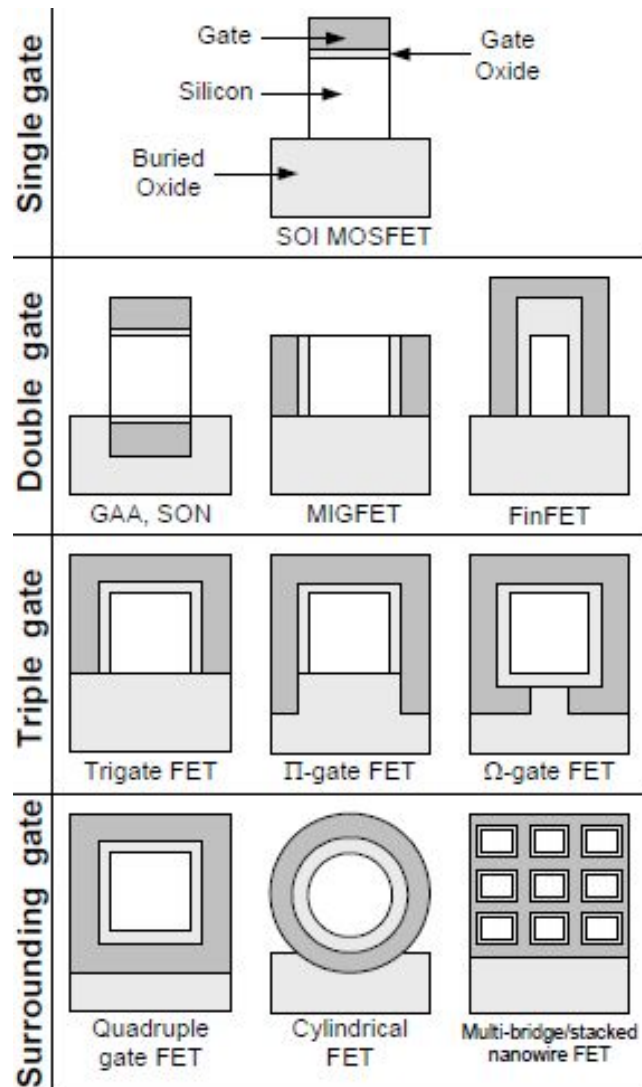
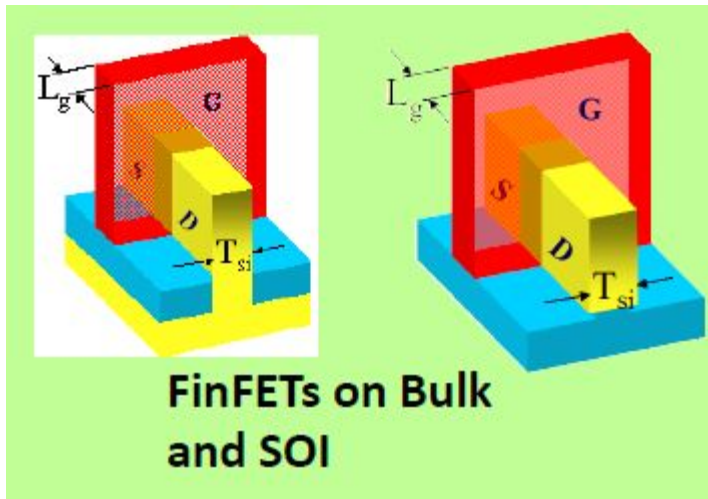
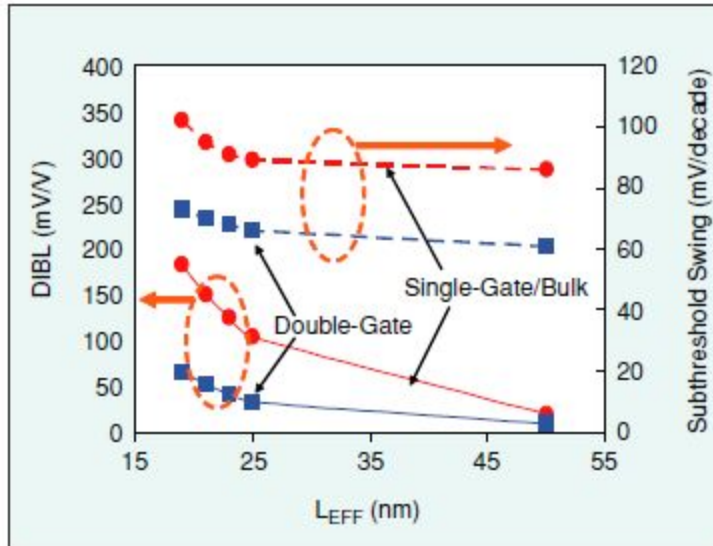


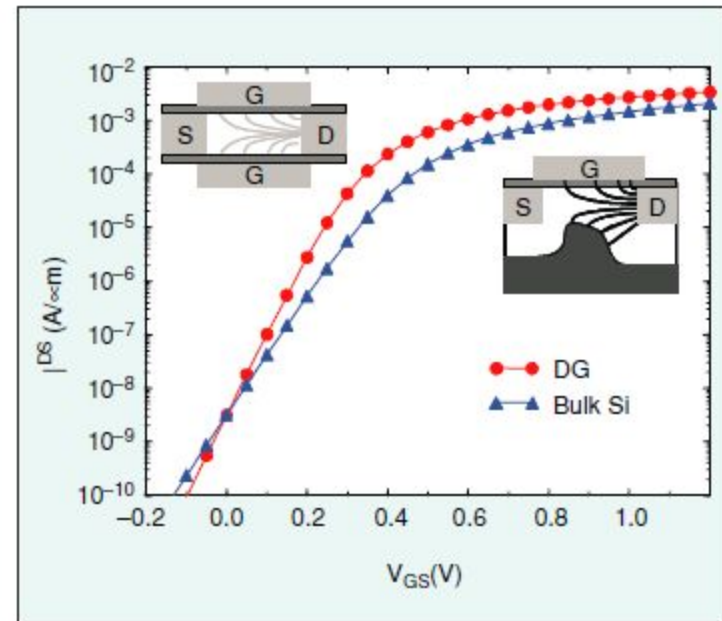
Fig. 1.9. Different gate structures.



# Single gate- double gate comparison



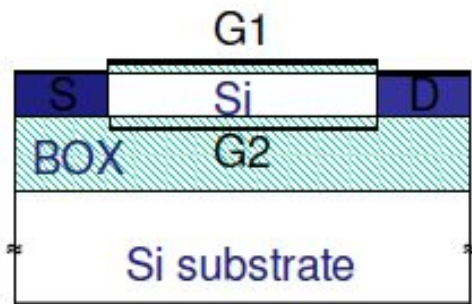
2. MEDICI-predicted DIBL and subthreshold swing versus effective channel length for DG and bulk-silicon nFETs. The DG device is designed with an undoped body and a near-mid-gap gate material.



3. Simulation of double-gate and single-gate FETs designed for equal subthreshold current density at  $V_{GS} = 0$  V illustrates the gain in drive current from improved channel control of the double-gate FET. Both gates contribute to control of the channel potential in subthreshold, while in the bulk case the gate must compete with the influence of the substrate.

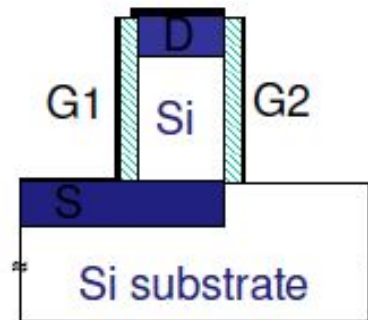
# Different ways to realize double gate

Planar DG MOSFET



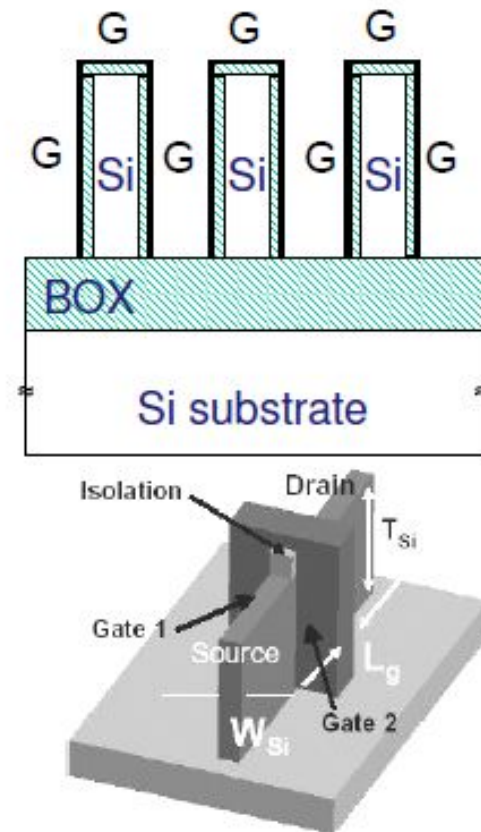
Problem of top and bottom gates alignment

Vertical MOSFET

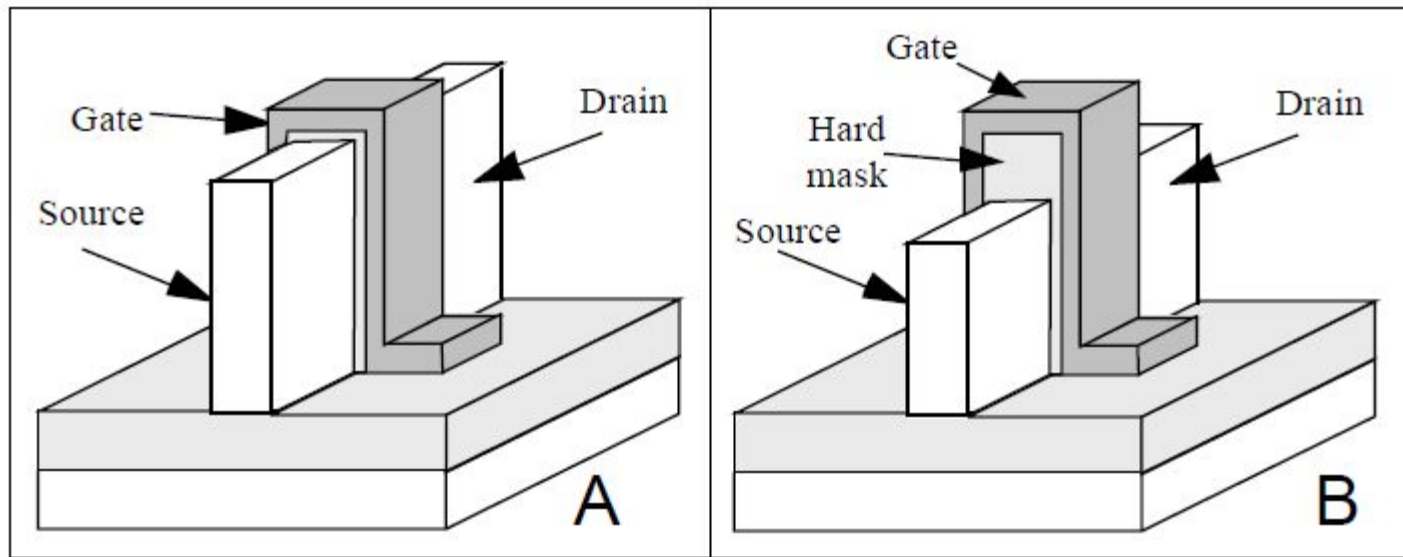


Increased overlap capacitance between G and S/D regions

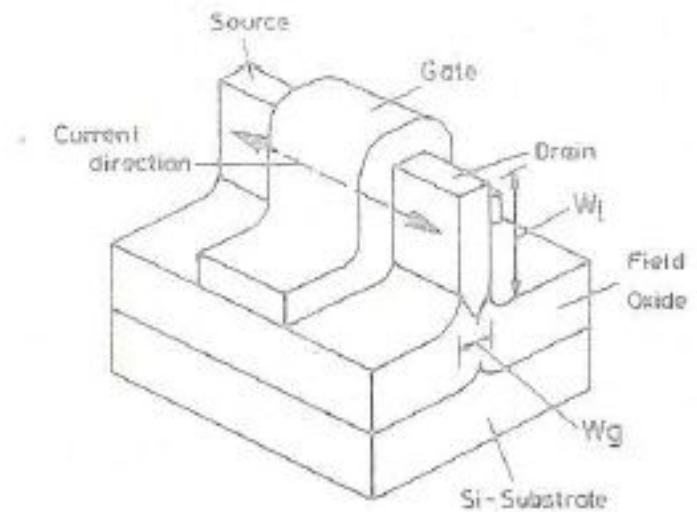
FinFET



# DELTA (Hitachi) vs FinFET (UC Berkeley)



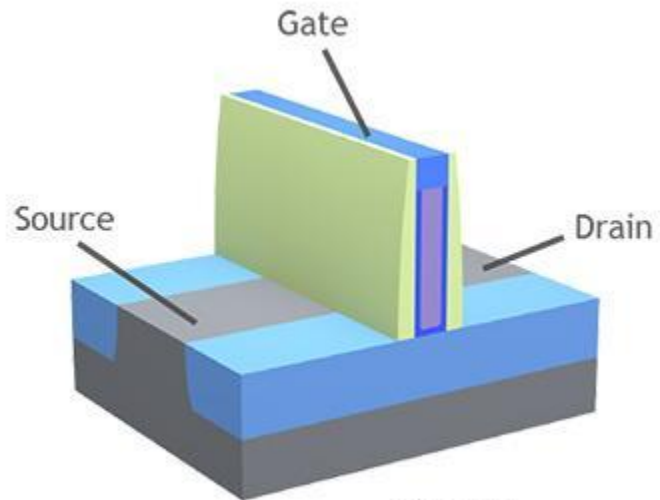
**Fig. 1.7.** Examples of double-gate MOS structure: A: DELTA MOSFET; B: FinFET.



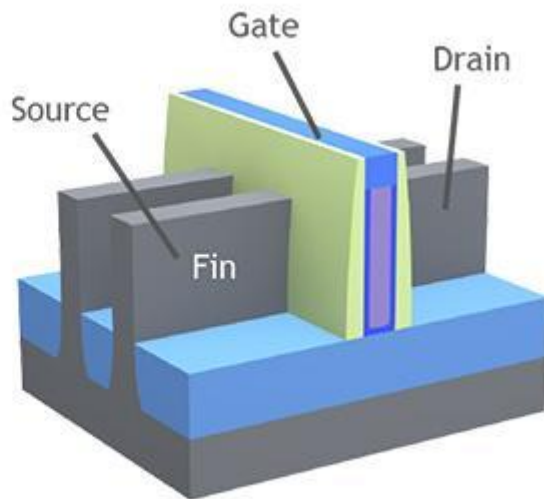
**FinFET was named after the dorsal fin of a fish**



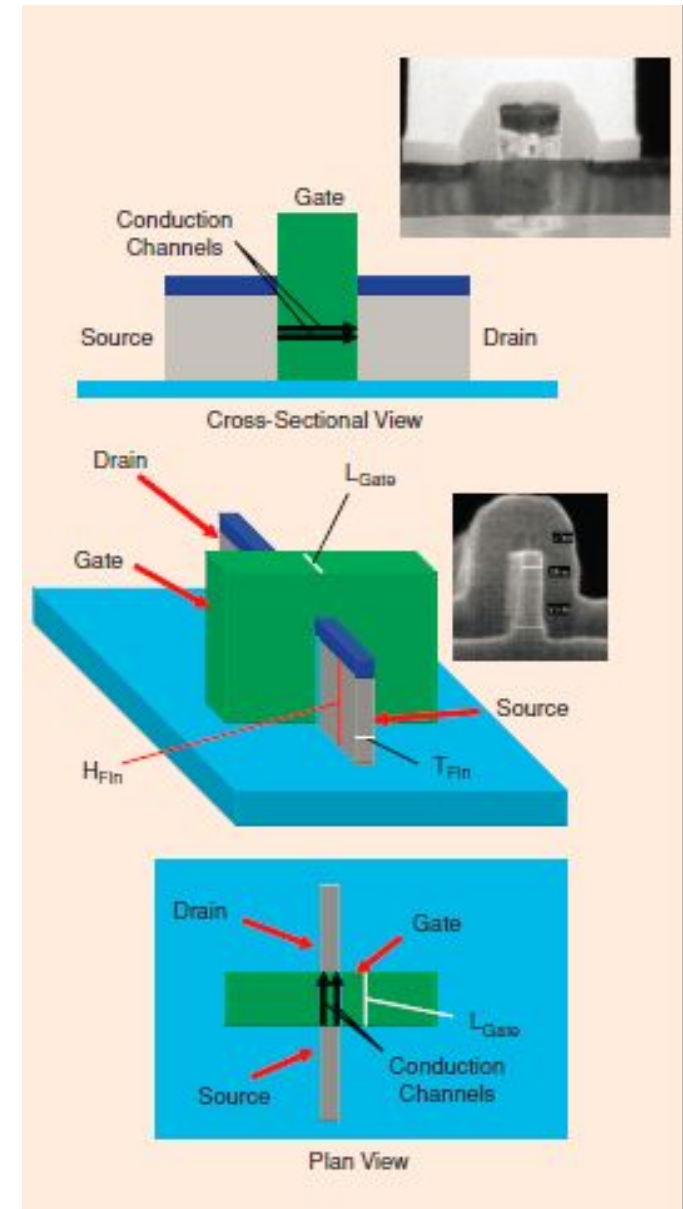
# FinFET cross section



Planar

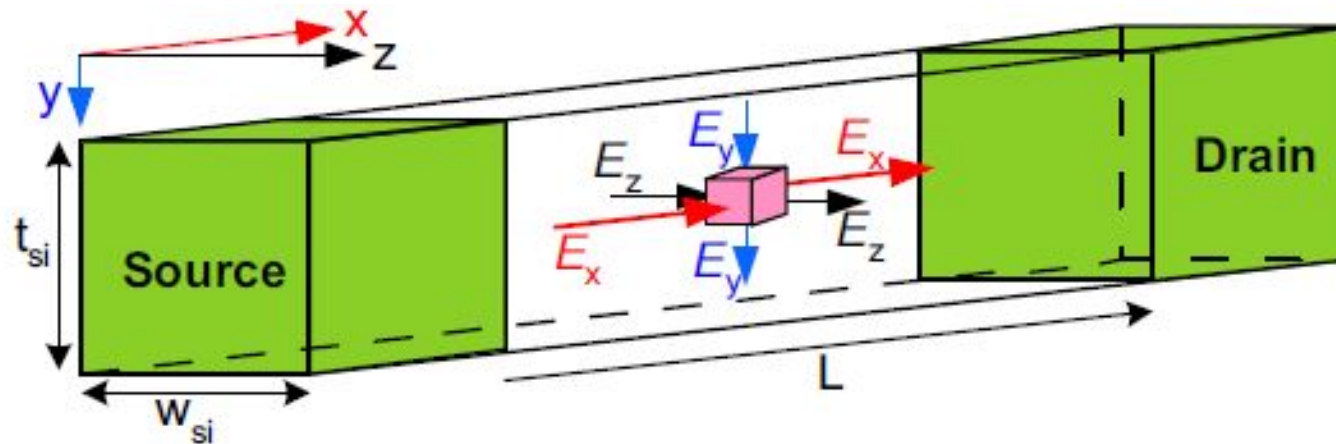


FinFET



Plan View

# Concept of natural length (not valid for bulk or PDSOI)



Poisson: 
$$\frac{d^2\Phi(x, y, z)}{dx^2} + \frac{d^2\Phi(x, y, z)}{dy^2} + \frac{d^2\Phi(x, y, z)}{dz^2} = -\frac{\rho(x, y, z)}{\epsilon_{si}}$$

Or, in other words: 
$$\frac{dE_x(x, y, z)}{dx} + \frac{dE_y(x, y, z)}{dy} + \frac{dE_z(x, y, z)}{dz} = C$$

In the case of a *wide* single- or double-gate device, we have  $\frac{d\Phi}{dz} = 0$ , and Poisson's equation becomes:

$$\frac{d^2\Phi(x, y)}{dx^2} + \frac{d^2\Phi(x, y)}{dy^2} = \frac{qN_a}{\epsilon_{si}} \quad (1.9)$$

$$\Phi(x, y) = c_0(x) + c_1(x)y + c_2(x)y^2 \quad (1.10)$$

#### 1.5.1.1.1 Single-gate SOI MOSFET

In the case of a single-gate SOI device the boundary conditions to equation 1.10 are:[102]

1.  $\Phi(x, 0) = \Phi_f(x) = c_0(x)$  where  $\Phi_f(x)$  is the front surface potential;

2.  $\left. \frac{d\Phi(x, y)}{dy} \right|_{y=0} = \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} = c_1(x)$

where  $\Phi_{gs} = V_{gs} - V_{FBF}$  is the front gate voltage,  $V_{gs}$ , minus the front-gate flat-band voltage,  $V_{FBF}$ .

3. If we assume that the buried oxide is very thick the potential difference across any finite distance in the BOX is negligible in the  $y$  direction we can write  $\frac{d\Phi(x, y)}{dy} \cong 0$  in the BOX. Therefore,

we have:  $\left. \frac{d\Phi(x, y)}{dy} \right|_{y=t_{si}} = c_1(x) + 2t_{si}c_2(x) \cong 0$  and thus  $c_2(x) \cong -\frac{c_1(x)}{2t_{si}}$ .

Introducing these three boundary conditions in equation 1.10 we obtain:

$$\Phi(x, y) = \Phi_f(x) + \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} y - \frac{1}{2t_{si}} \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} y^2 \quad (1.11)$$

Substituting equation 1.11 into equation 1.9, and setting  $y=0$ , at which depth  $\Phi(x, y) = \Phi_f(x)$  we obtain:

$$\frac{d^2\Phi_f(x)}{dx^2} - \frac{\varepsilon_{ox}}{\varepsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{si} t_{ox}} = \frac{qN_a}{\varepsilon_{si}} \quad (1.12)$$

Once  $\Phi_f(x)$  is determined from equation 1.12,  $\Phi(x, y)$  can be calculated using equation 1.11. Equation 1.12, however, can be used for another purpose. If we define:

$$\lambda_1 = \sqrt{\frac{\varepsilon_{si}}{\varepsilon_{ox}} t_{ox} t_{si}} \quad (1.13)$$

and

$$\varphi(x) = \Phi_f(x) - \Phi_{gs} + \frac{qN_a}{\varepsilon_{si}} \lambda_1^2 \quad (1.14)$$

then equation 1.12 can be re-written as follows:

$$\frac{d^2\varphi(x)}{dx^2} - \frac{\varphi(x)}{\lambda_1^2} = 0 \quad (1.15)$$

This equation has a solution in the form  $\varphi(x) = \varphi_0 \exp\left(\pm \frac{x}{\lambda_1}\right)$  where  $\lambda_1$

### 1.5.1.1.2 Double-gate SOI MOSFET

In the case of a double-gate device the boundary conditions to equation 1.10 are:

1.  $\Phi(x,0) = \Phi(x,t_{si}) = \Phi_f(x) = c_o(x)$  where  $\Phi_f(x)$  is the front surface potential;

- 
2.  $\left. \frac{d\Phi(x,y)}{dy} \right|_{y=0} = \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} = c_1(x)$  where  $\Phi_{gs} = V_{gs} - V_{FBF}$  is the front gate voltage,  $V_{gs}$ , minus the front gate flat-band voltage,  $V_{FBF}$ .

3.  $\left. \frac{d\Phi(x,y)}{dy} \right|_{y=t_{si}} = -\frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} = c_1(x) + 2t_{si} c_2(x) = -c_1(x)$  and  
thus  $c_2(x) = \frac{-c_1(x)}{t_{si}}$ .

Substituting these boundary conditions into equation (1.15) yields:

$$\Phi(x,y) = \Phi_f(x) + \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} y - \frac{1}{t_{si}} \frac{\epsilon_{ox}}{\epsilon_{si}} \frac{\Phi_f(x) - \Phi_{gs}}{t_{ox}} y^2 \quad (1.16)$$

The key difference between this expression and equation 1.10 is that the term  $1/2t_{si}$  has now been replaced by  $1/t_{si}$ . In other words it looks as if the double-gate device was twice as thin as the single-gate transistor. The natural length of the double-gate device can be derived the same way it was done for the single-gate case, which yields:

$$\lambda_2 = \sqrt{\frac{\epsilon_{si}}{2\epsilon_{ox}} t_{ox} t_{si}} \quad (1.17)$$

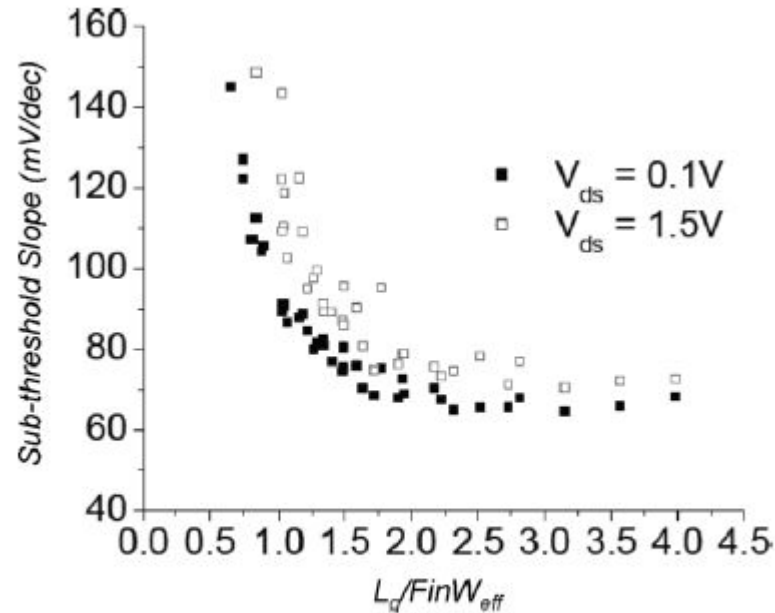
**Table 1.2.** Natural length in devices with different geometries.

|                                                      |                                                                                                                                          |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Single gate                                          | $\lambda_1 = \sqrt{\frac{\epsilon_{si}}{\epsilon_{ox}}} t_{si} t_{ox}$                                                                   |
| Double gate                                          | $\lambda_2 = \sqrt{\frac{\epsilon_{si}}{2\epsilon_{ox}}} t_{si} t_{ox}$                                                                  |
| Quadruple gate<br>(square channel cross section)     | $\lambda_4 \cong \sqrt{\frac{\epsilon_{si}}{4\epsilon_{ox}}} t_{si} t_{ox}$                                                              |
| Surrounding gate<br>(circular channel cross section) | $\lambda_o = \sqrt{\frac{2\epsilon_{si} t_{si}^2 \ln\left(1 + \frac{2t_{ox}}{t_{si}}\right) + \epsilon_{ox} t_{si}^2}{16\epsilon_{ox}}}$ |

Short channel effects can be reduced by reducing silicon film thickness also. By using more gate one can relax the requirements on silicon film thickness, compared to single gate.

**Min feature size in MuG technology is fin width and not gate length**

# Fin width requirements



**Fig. 2.4.** Measured subthreshold swing vs.  $L_g/FinW_{eff}$ .  $FinWIDTH$  varies from 20nm to 90nm;  $L_g$  varies from 30nm to 120nm. The subthreshold slope is shown for both for  $V_d=0.1V$  and 1.5V.[22] Copyright© 2001 IEEE.

$L_G/Fin_{Width} \geq 1.5$  for good electrostatics

# Fin height requirements

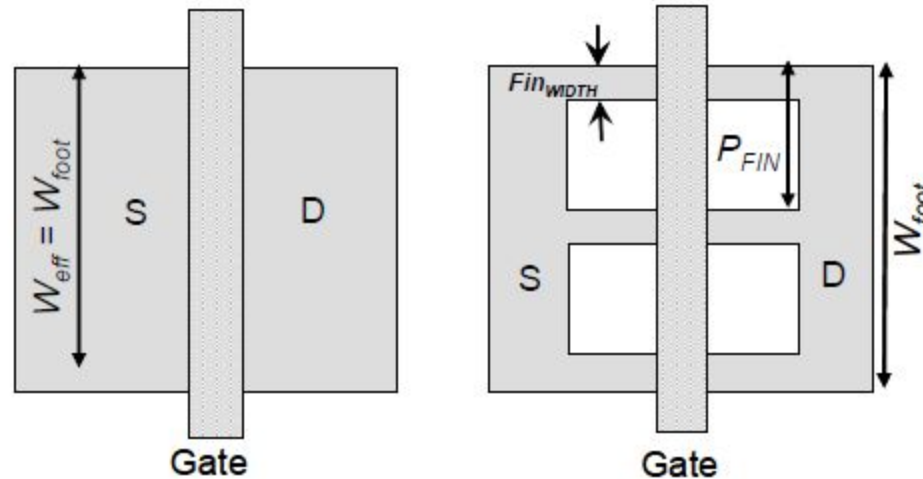


Fig. 2.5. Layout comparison between a planar MOSFET and a typical FinFET.

$$W_{\text{eff}} = 2 \text{Fin}_{\text{HEIGHT}} \frac{W_{\text{foot}}}{P_{\text{FIN}}} \quad (2.14)$$

If we want the FinFET layout to be "competitive", we need  $W_{\text{eff}} \geq W_{\text{foot}}$ , or:

$$2 \text{Fin}_{\text{HEIGHT}} \geq P_{\text{FIN}} \quad (2.15)$$

Quantization of width: available in multiples of  $2 \cdot \text{Fin}_{\text{Height}}$

$\text{Fin}_{\text{Height}}$  limited by stability and gate etch capability

# FIN xTEMs

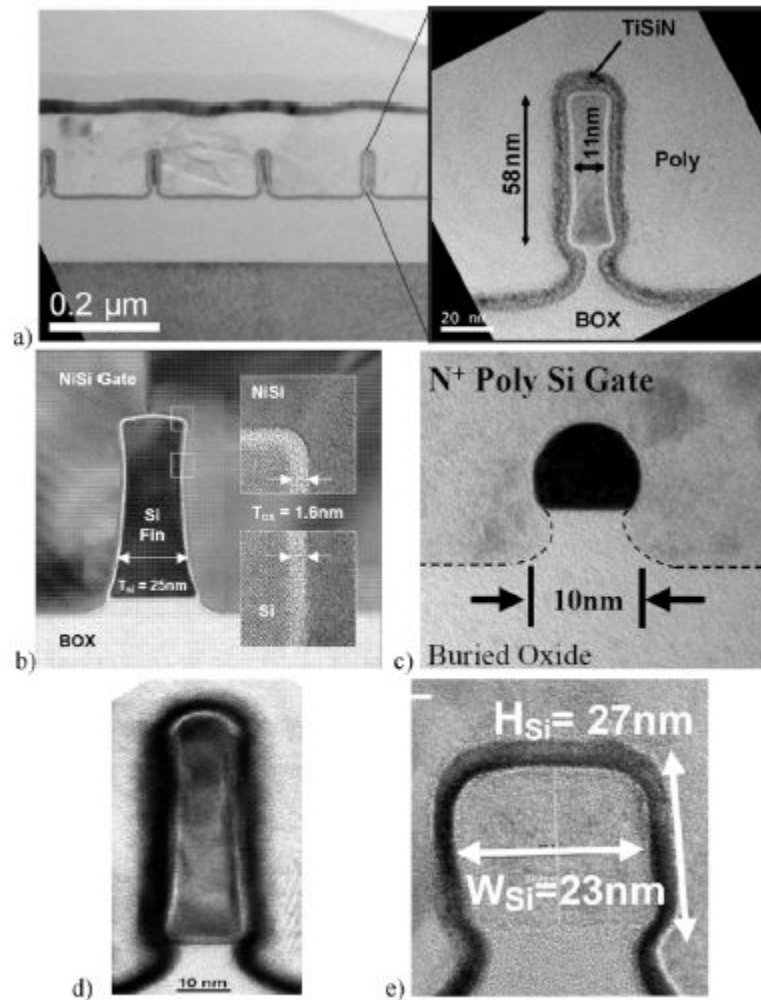
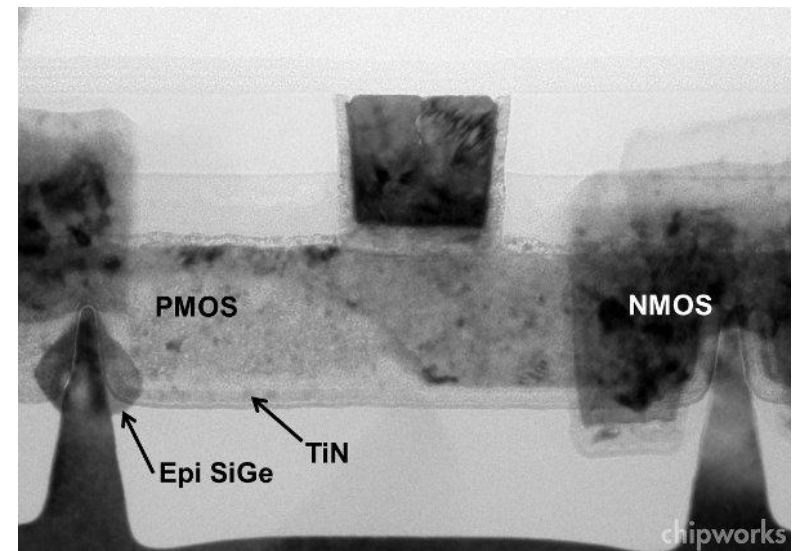
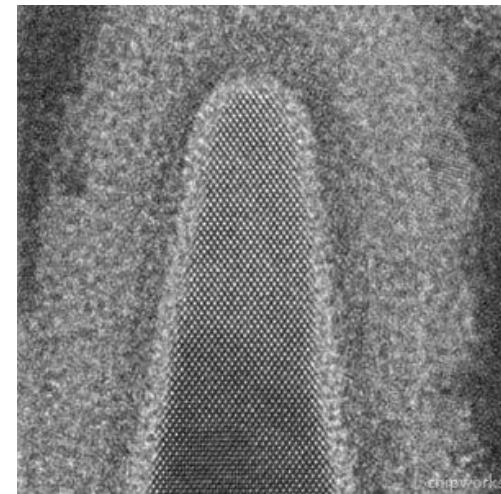
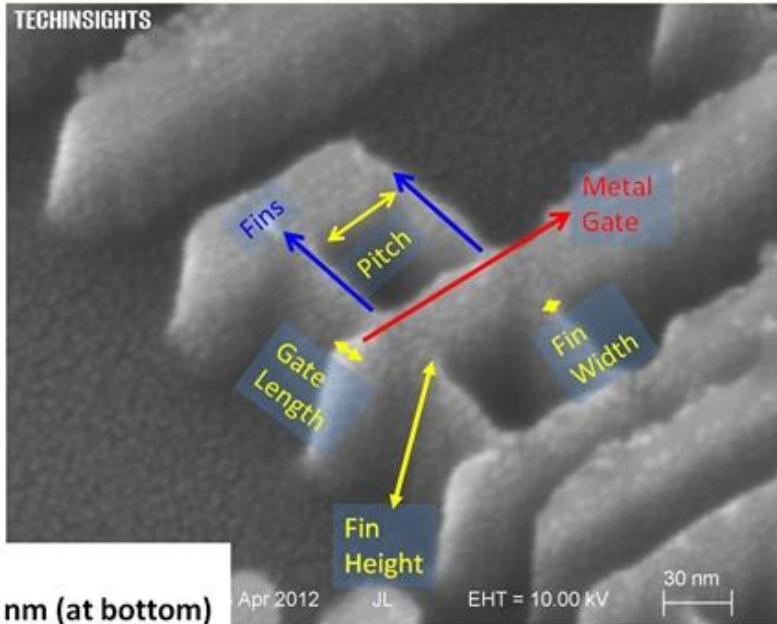
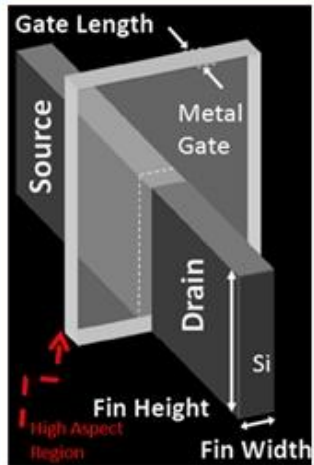


Fig. 2.7. Cross-sectional TEM of fins under gates. a) [27] Copyright© 2006 IEEE, b) [28] Copyright© 2002 IEEE, c) [29] Copyright© 2004 IEEE, d) [30] Copyright© 2005 IEEE, e) [31] Copyright© 2006 IEEE.

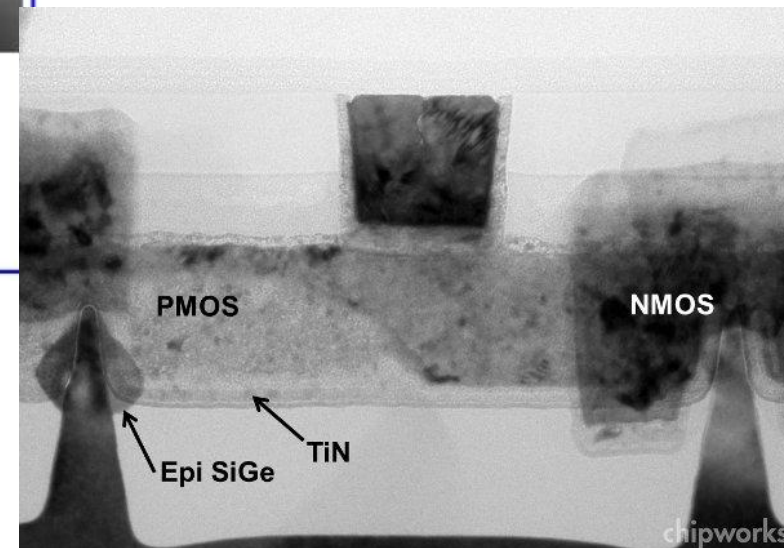
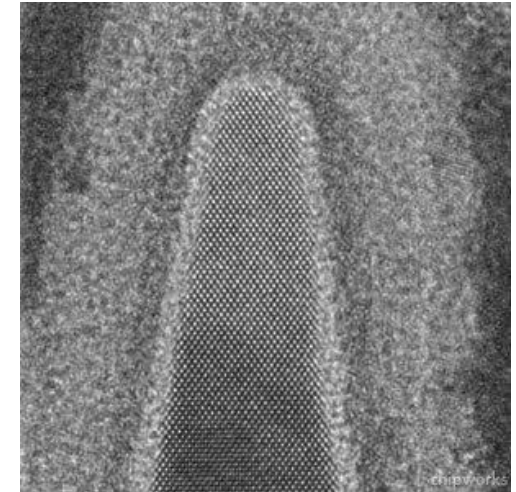


# Intel 22nm technology



Average Gate height:  
 Average Fin width: 18 nm (at bottom)  
 Average Fin height: 35 nm  
 Average Fin pitch: 60 nm

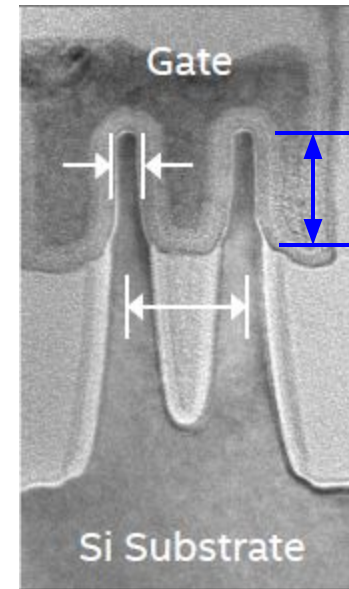
Average Tri-Gate length: ~ 30 nm  
 Average Tri-Gate width: ~ 80 nm



Intel 22nm Trigate xTEM  
 Ref: Chipworks

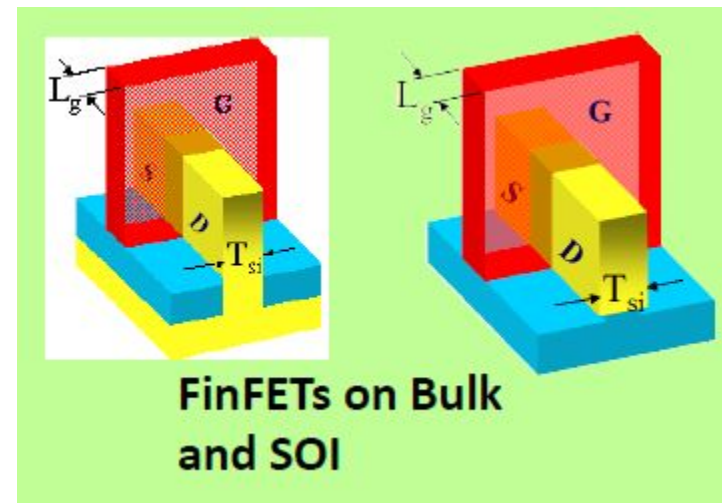
# Intel 14nm technology (2014)

8 nm Fin width  
 42 nm Fin pitch  
 42 nm Fin height  
 52 nm M1 pitch



Width of transistor =  $(2 * \text{Fin height} + \text{Width of fin}) * \text{No: of fins}$

|                       | 22 nm<br>Node | 14 nm<br>Node | Scale |
|-----------------------|---------------|---------------|-------|
| Transistor Fin Pitch  | 60            | 42            | .70x  |
| Transistor Gate Pitch | 90            | 70            | .78x  |
| Interconnect Pitch    | 80            | 52            | .65x  |
|                       | nm            | nm            |       |



# Intel 10nm technology (1H 2021)

7 nm Fin width  
 34 nm Fin pitch  
 46 nm Fin height  
 36 nm M1 pitch

|                       | 22 nm<br>Node | 14 nm<br>Node | Scale |
|-----------------------|---------------|---------------|-------|
| Transistor Fin Pitch  | 60            | 42            | .70x  |
| Transistor Gate Pitch | 90            | 70            | .78x  |
| Interconnect Pitch    | 80            | 52            | .65x  |
|                       | nm            | nm            |       |

|      | Intel<br>First Production |
|------|---------------------------|
| 1999 | 180 nm                    |
| 2001 | 130 nm                    |
| 2003 | 90 nm                     |
| 2005 | 65 nm                     |
| 2007 | 45 nm                     |
| 2009 | 32 nm                     |
| 2011 | 22 nm                     |
| 2014 | 14 nm                     |
| 2016 | <del>10 nm</del>          |
| 2017 | <del>10 nm</del>          |
| 2018 | <del>10 nm?</del>         |
| 2019 | <del>10 nm!</del>         |

**Finally in 2021**

# Sidewall Image transfer (SIT) lithography



**SOI wafer**



**Grow PadOx**



**Deposit Mandrel**



**Pattern Mandrel**



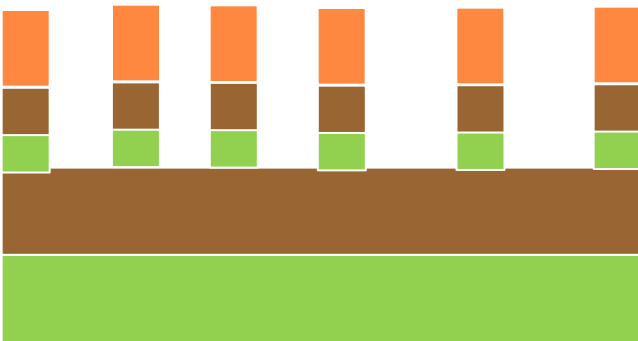
**Deposit Nitride Spacer**



**After spacer etch**



**After removing mandrel**

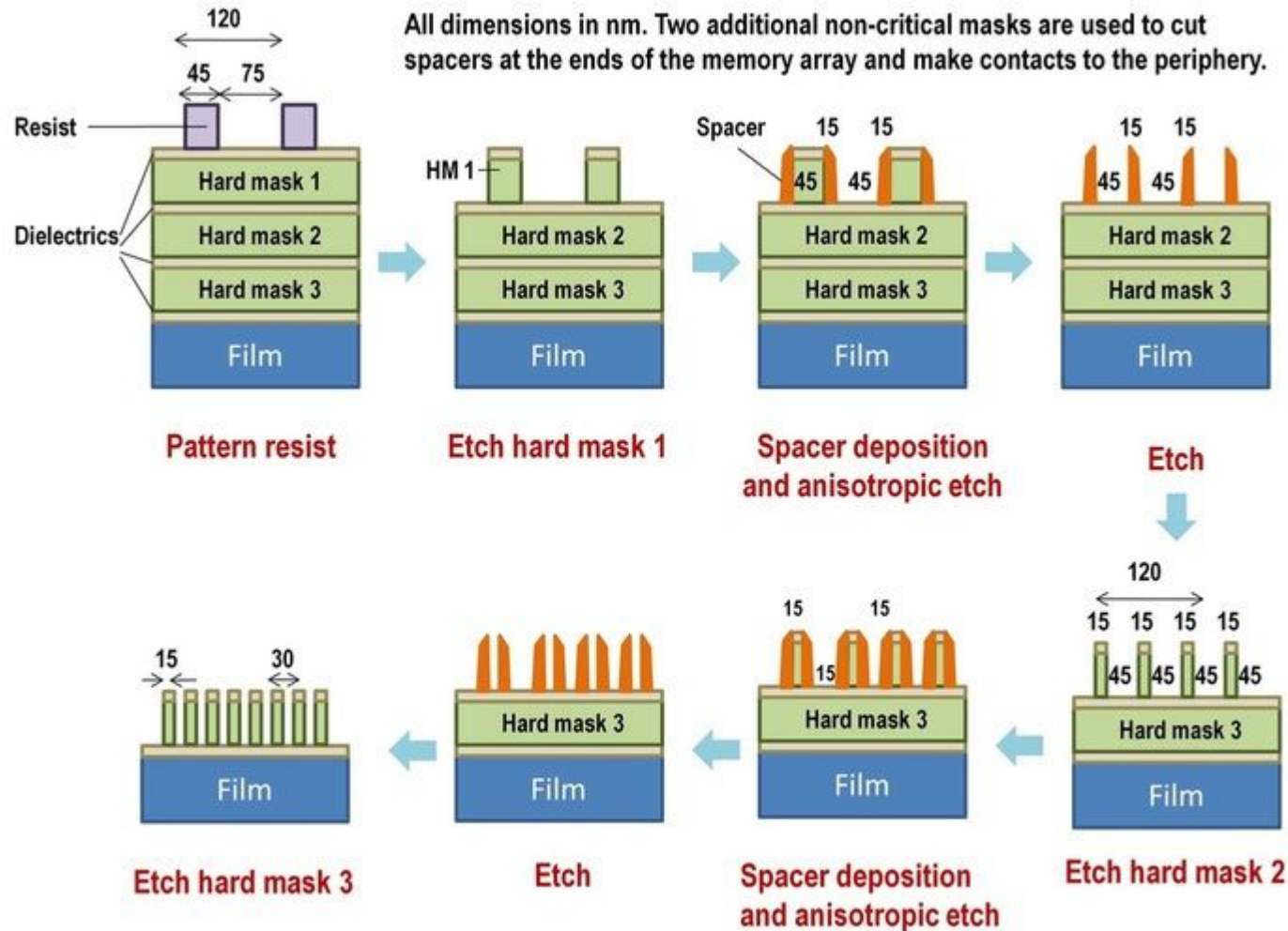


**After SacOx & Si RIE**

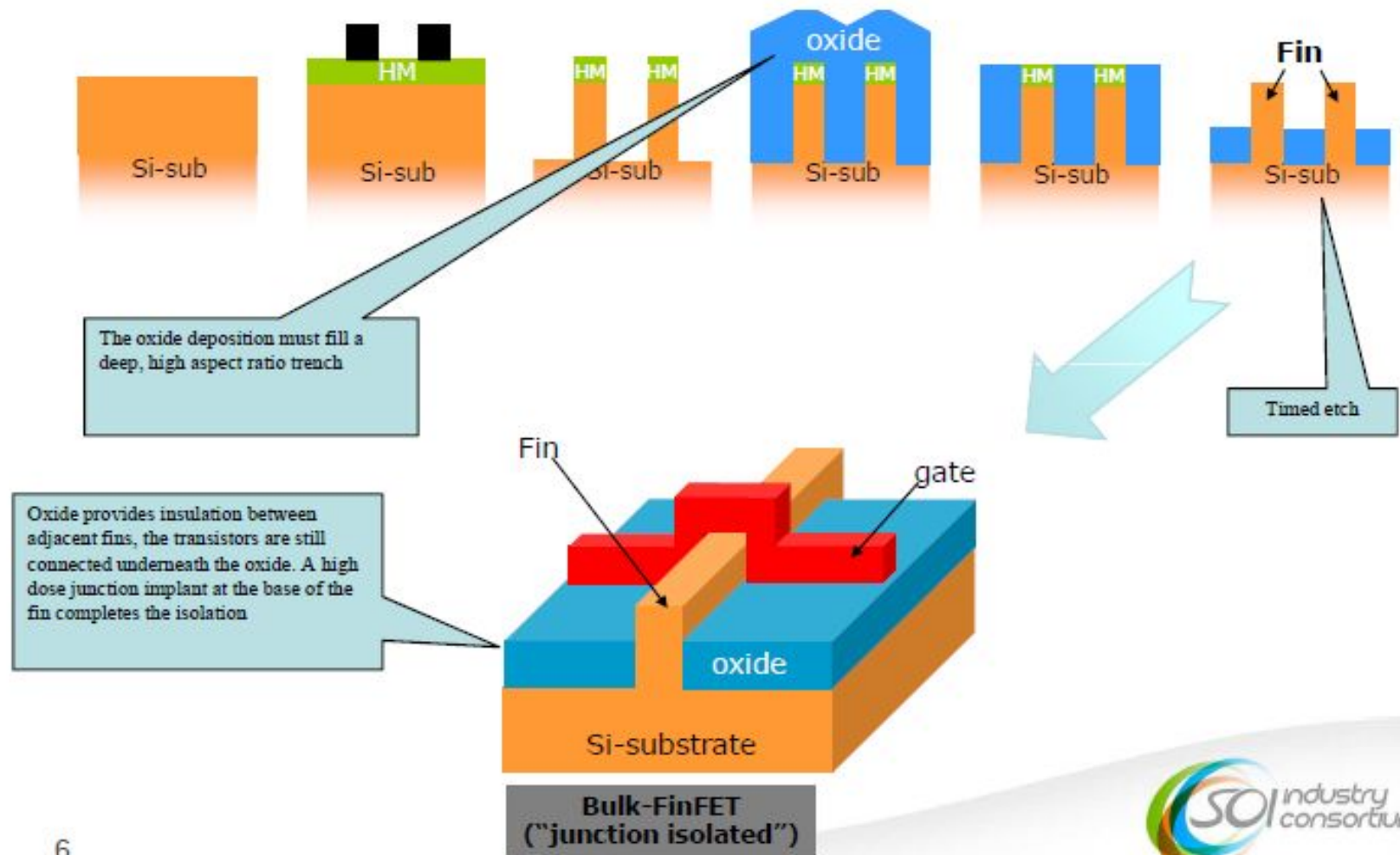


**After Nitride removal**

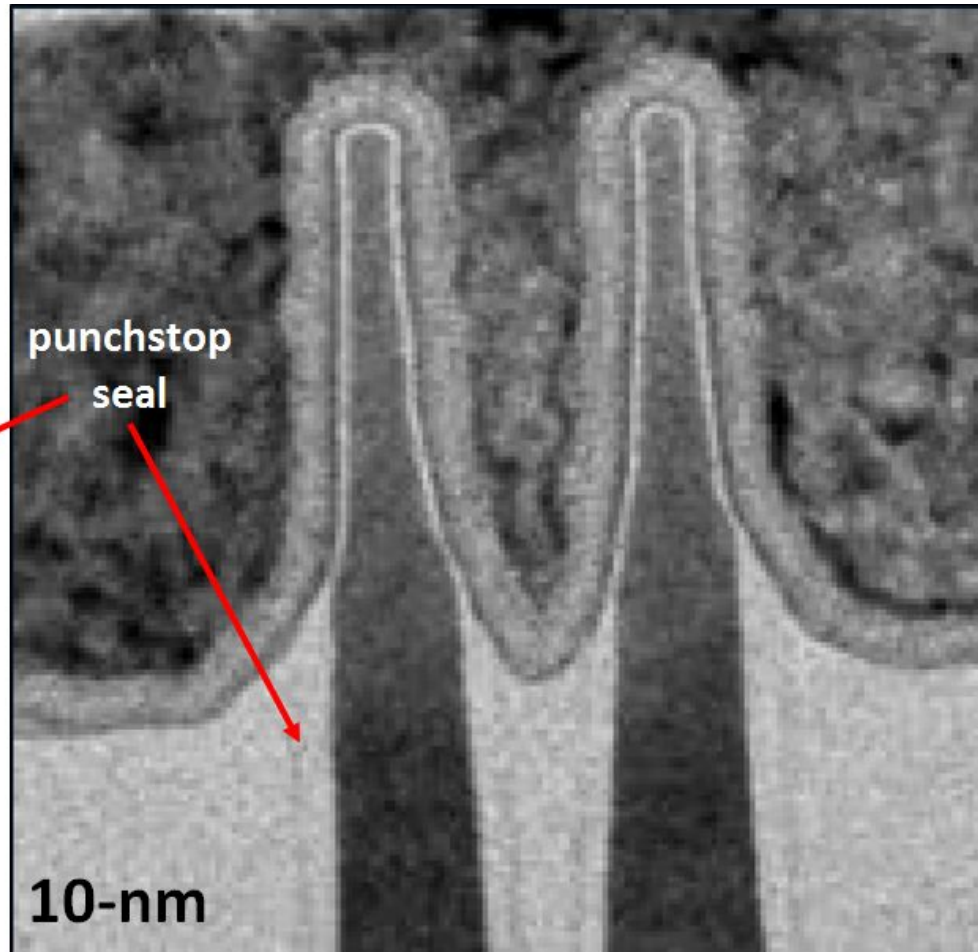
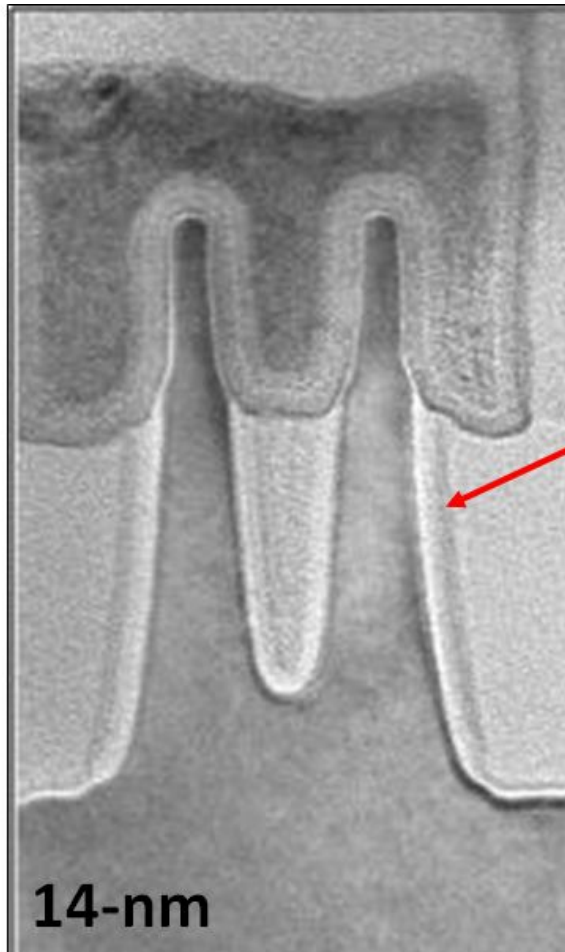
# Self-aligned quad patterning (SAQP)



## Junction Isolated Bulk FinFET flow

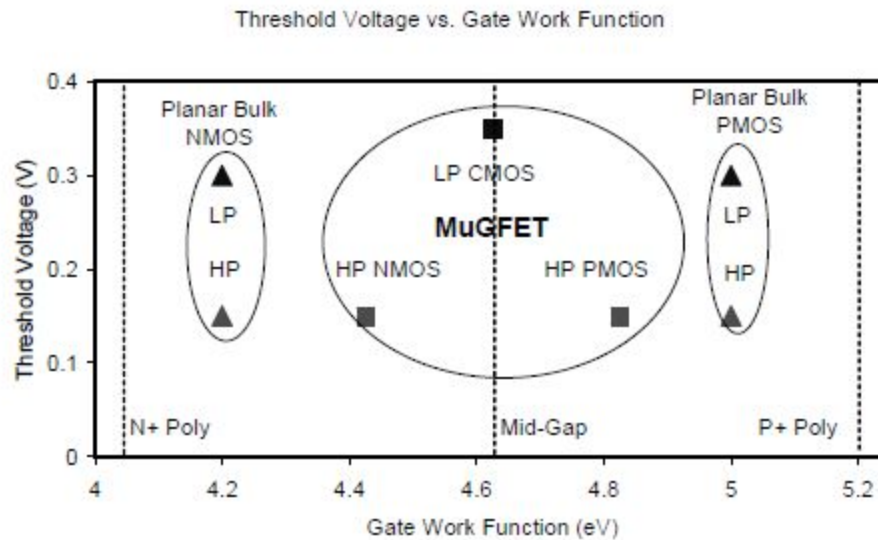


# Solid phase diffusion for punchstopper

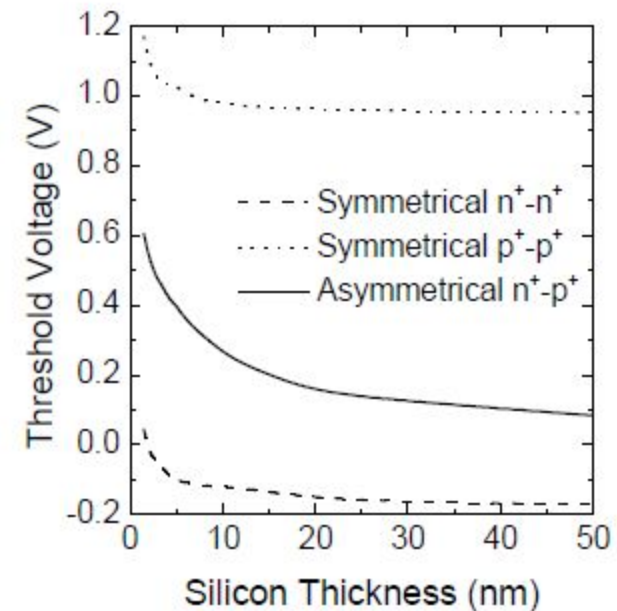


Source: Intel

# Gate work function requirement



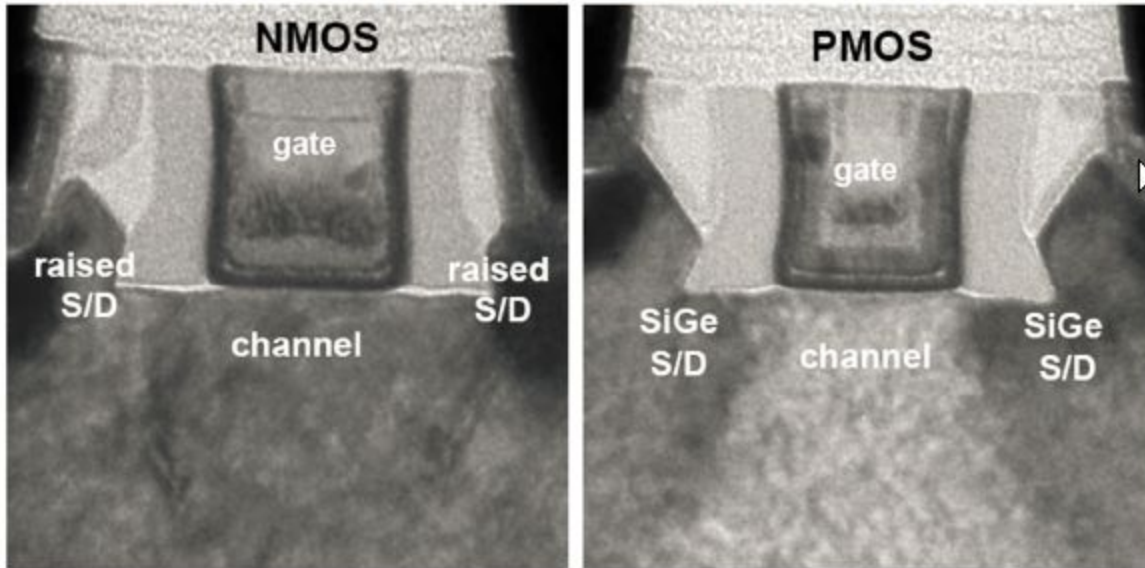
**Fig. 2.20.** Gate workfunctions for different device types. High Performance (HP) and Low Power (LP) planar bulk MOSFETs need workfunctions close to the silicon band edges, while HP and LP multi-gate MOSFETs need workfunction values close to the midgap of the silicon.



**Fig. 5.17.** Threshold voltage versus silicon thickness for different DGSOI structures.

# Epitaxial Raised S/D (Intel 32nm)

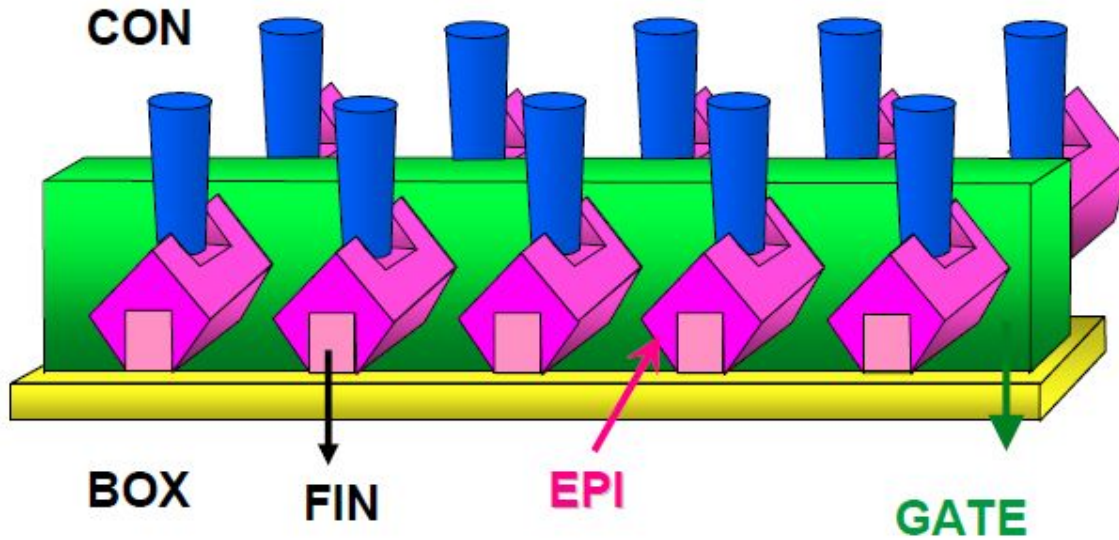
Cross-sectional TEM views of Intel's 32 nm CMOS devices



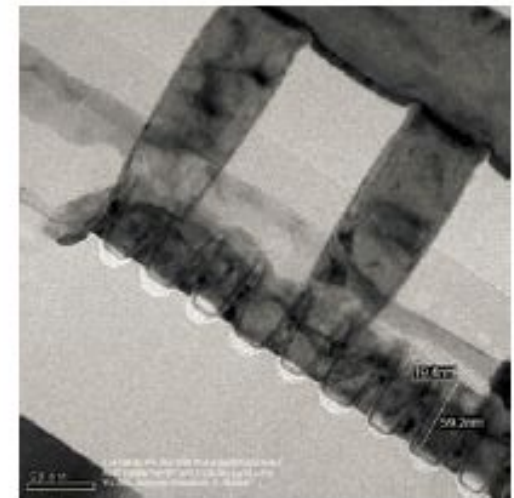
P. Packan et al. (Intel), *IEDM Technical Digest*, pp. 659-662, 2009

**To reduce series resistance**

# Merged Epitaxial S/D



Merged epitaxy



# FinFET process flow

1. Fin formation
2. Poly Si dep and dummy gate Stack patterning
3. Spacer1 dep and removal of spacer from FIN areas
4. Epitaxial raised source/drain and merging fins
5. ILD deposition/RMG
6. Trench Contact (Tungsten plug) Formation
7. Inter layer dielectric
8. First Metal layer

**Removal of gate material between the fins and spacer on the fins is critical here**

# Controlling Threshold Voltage with Work Function Metals

33

NVT



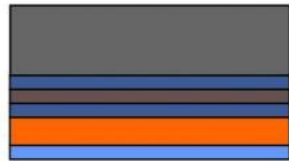
PVT



- Form interfacial oxide
- Deposit high-k gate oxide
- Deposit TiN cap
- Deposit TaN etch stop



- Deposit TiN
- Mask and etch off TiN



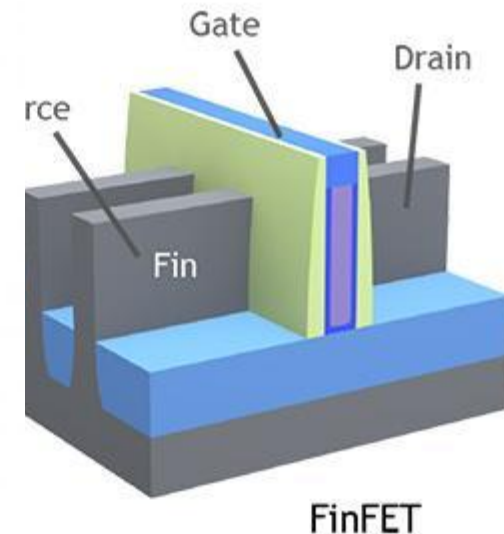
- Deposit TiAlC fill



NVT

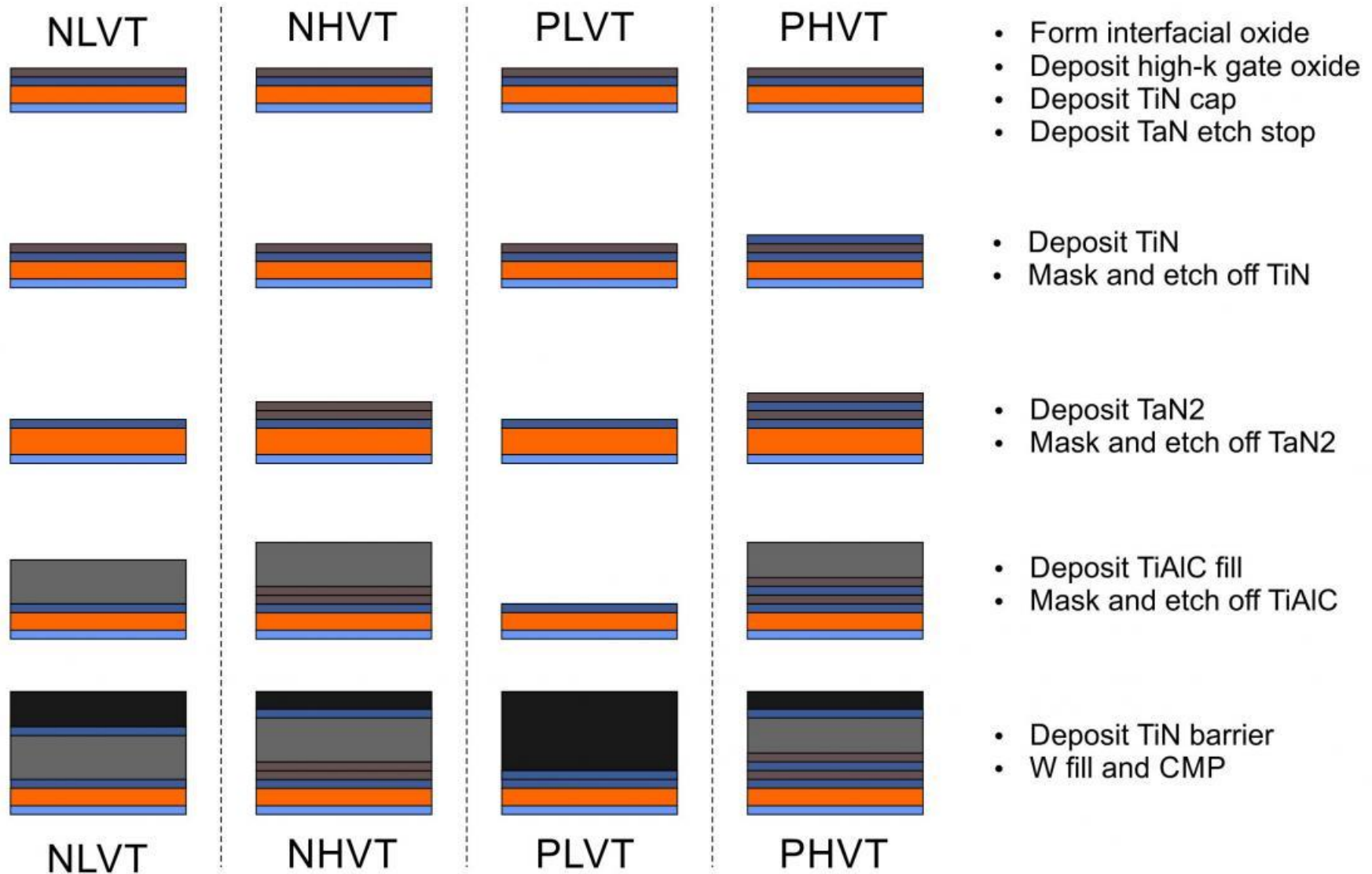
PVT

- Deposit TiN barrier
- W fill and CMP

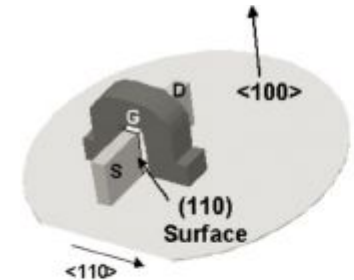
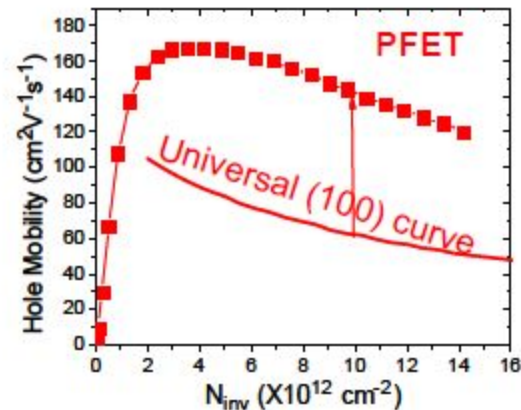
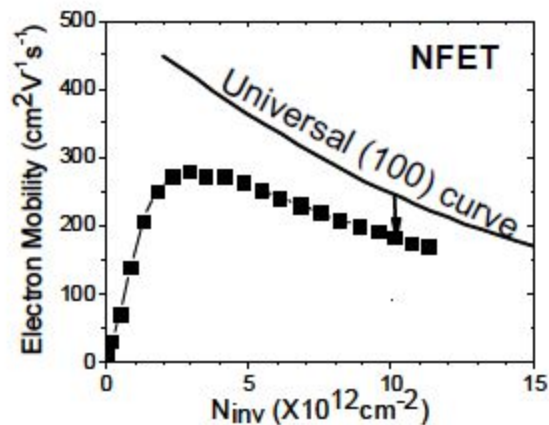


# Extending to 2 V<sub>T</sub>s

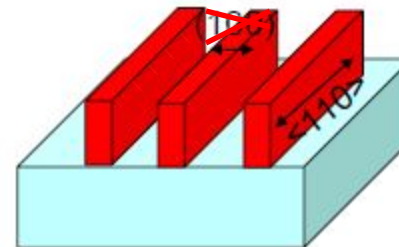
34



# FinFET orientation

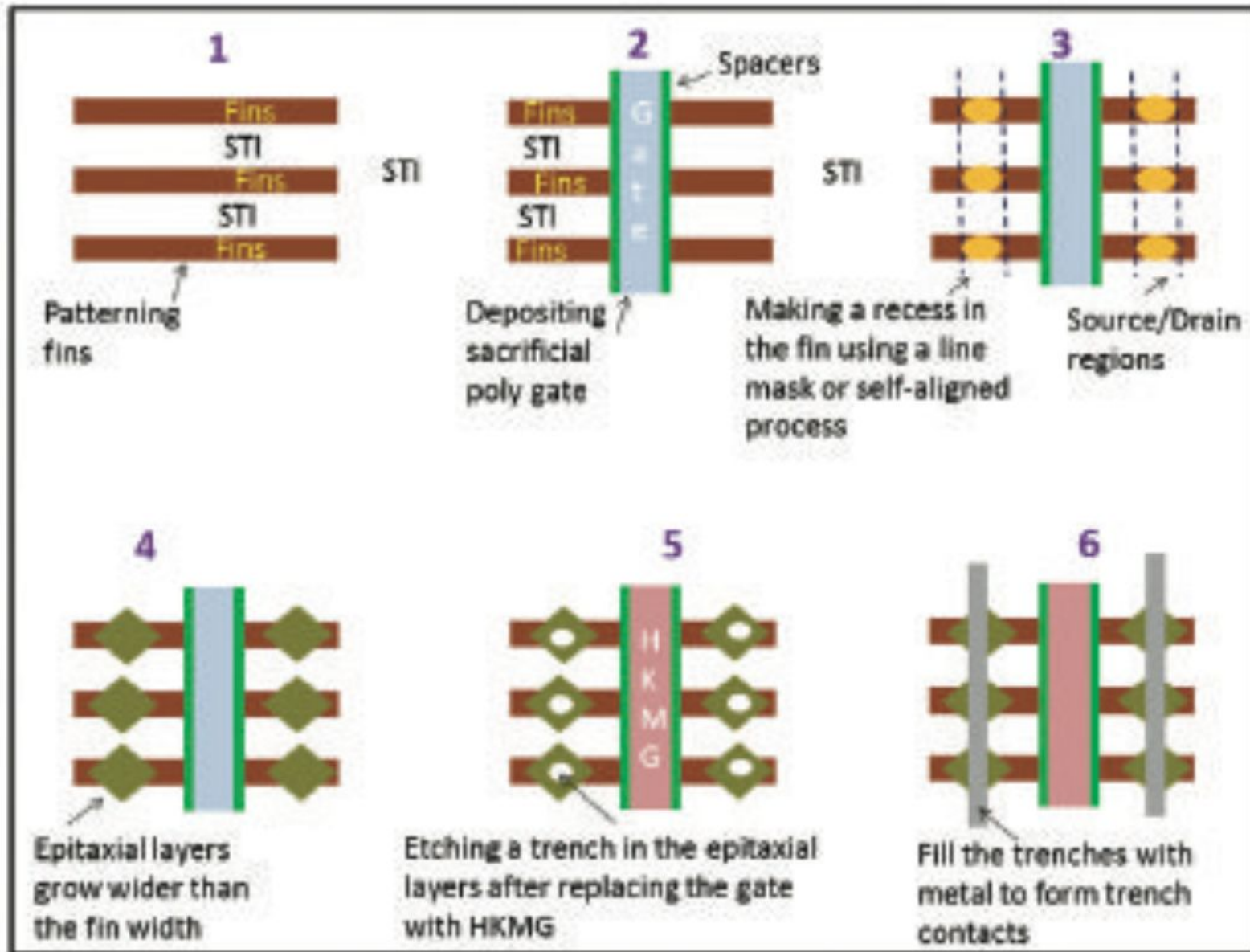


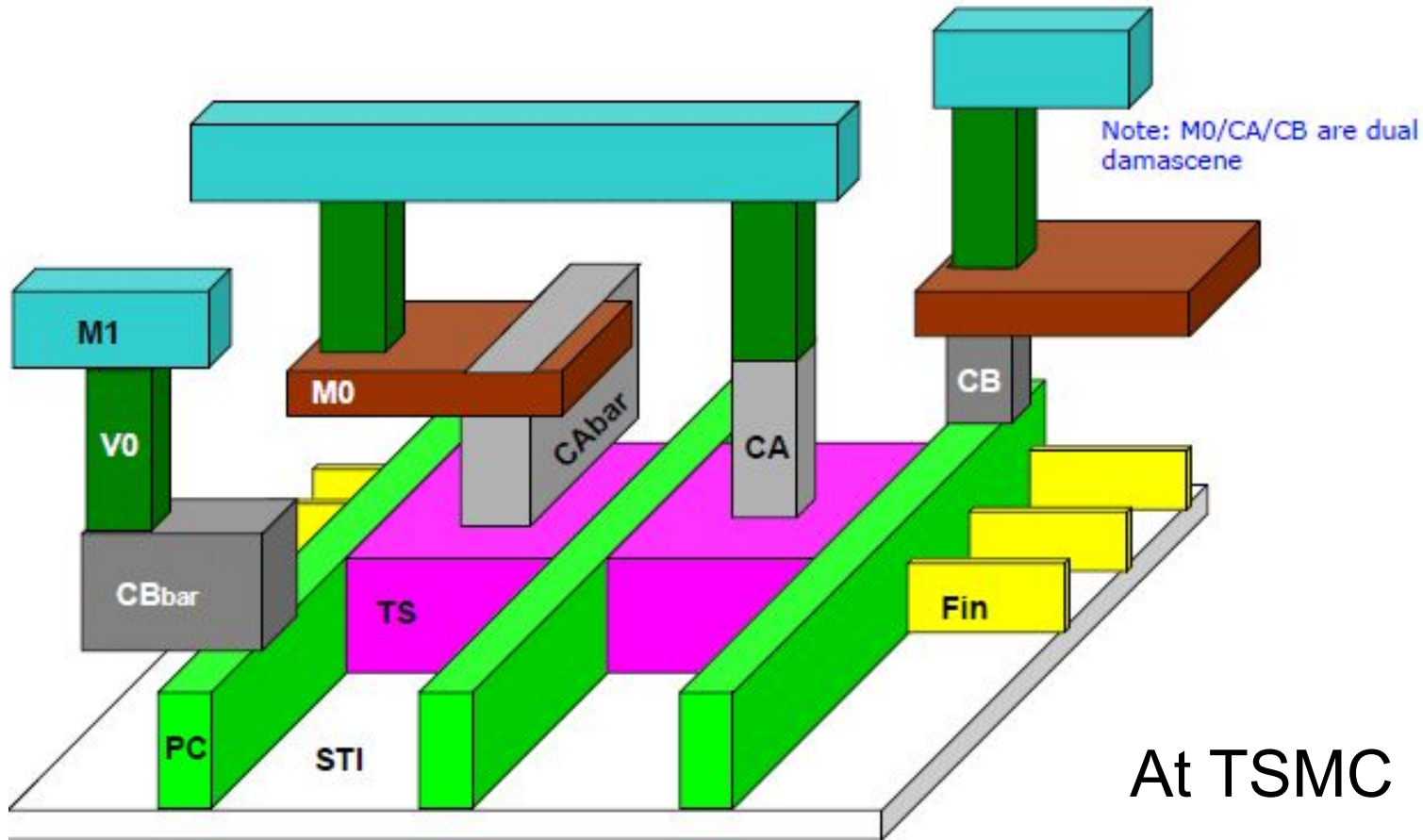
- Channel is formed on  $\langle 110 \rangle$  surface when fins are formed from (100) substrate
- PFET mobility improves significantly
  - $\sim 2\times$  improvement as expected
- Measurements show NFET mobility not degraded significantly
  - NFET degradation is  $< 20\%$



V. Basker, IBM, VLSI 2011

# MOL module - Trench contacts (14nm and below) 36





TS=MD  
CA=VD  
CB=VG

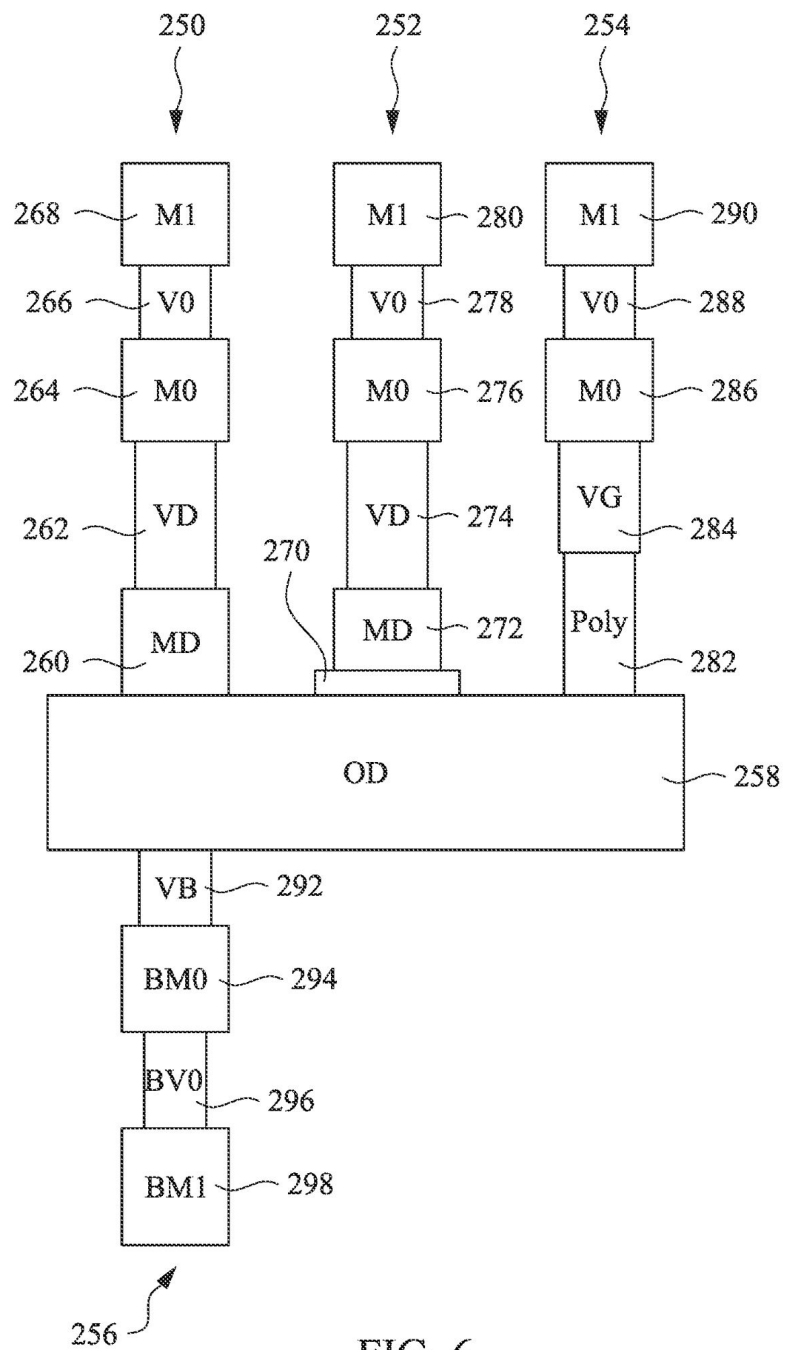
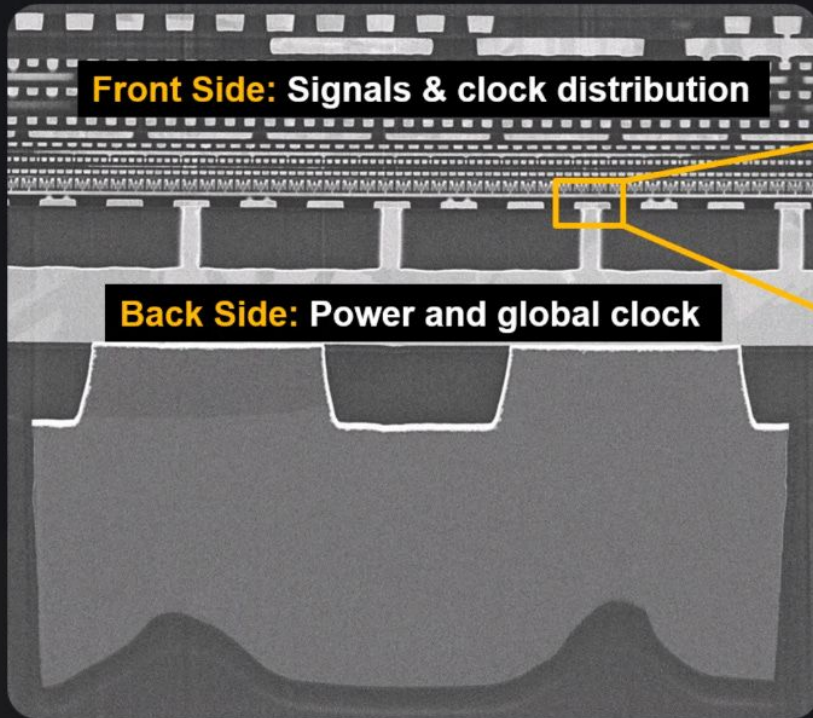


FIG. 6

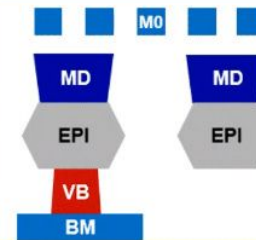
# TSMC A16™ Novel Backside Power Delivery Solution



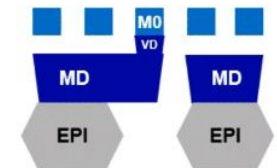
Unleash Innovation



**Innovative Backside Power Contact**



**Conventional Power Contact**



# Self-consistent solution

FET). As a result, energy subbands are formed. The  $j$ -th electron wave function and the corresponding energy level  $E_j$  can be found solving the Schrödinger equation using the effective mass approximation:

$$\left( -\frac{\hbar^2}{2m^*} \nabla^2 - q\Phi \right) \Psi_j = E_j \Psi_j \quad (1.21)$$

and the Poisson equation:

$$\nabla^2 \Phi = -\frac{q}{\epsilon_{si}} [p - n + N_D - N_A] \quad (1.22)$$

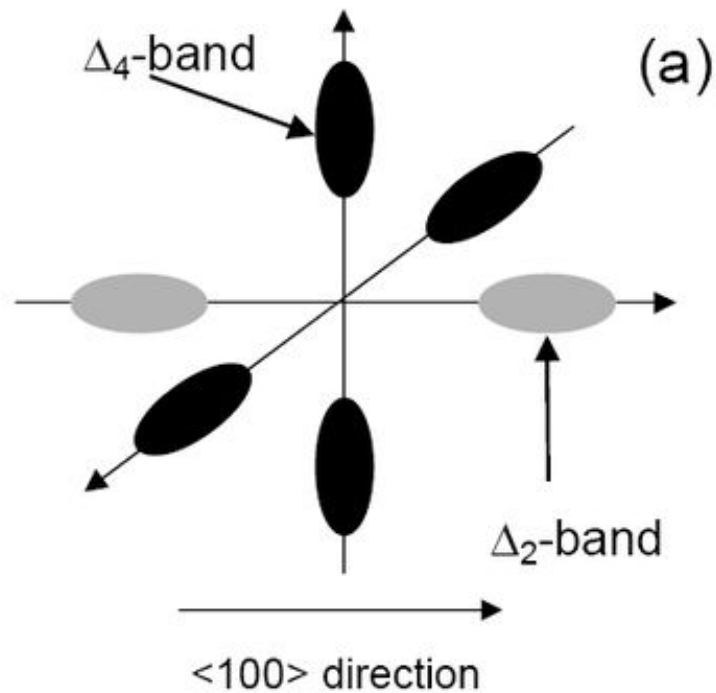
in a self-consistent manner. The electron concentration being given by:

$$n = \sum_j \left[ (\Psi_j \times \Psi_j^*) \times \int_{E_j}^{\infty} \rho_j(E) f_{FD}(E) dE \right] \quad (1.23)$$

where  $\rho_j(E)$  is the density of states as a function of the energy and  $f_{FD}(E)$  is the Fermi-Dirac distribution function. In a 2DEG the density of states is a

$$n(\mathbf{r}) = \sum_{j=1,2} \sum_i \frac{g_{vj} m_{d2Dj}}{\pi \hbar^2} k_B T \ln \left[ 1 + \exp \left( \frac{E_{Fn} - E_{ij}}{k_B T} \right) \right] |\varphi_{ij}(z)|^2$$

# Band structure (recap)

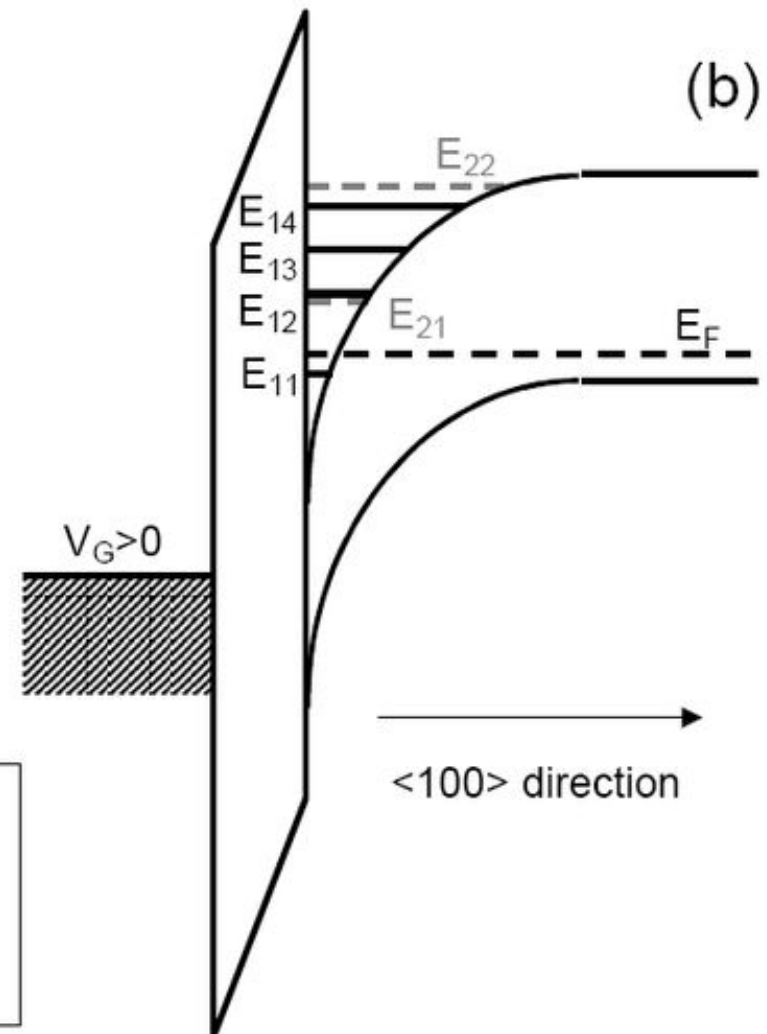


$\Delta_2$ -band:

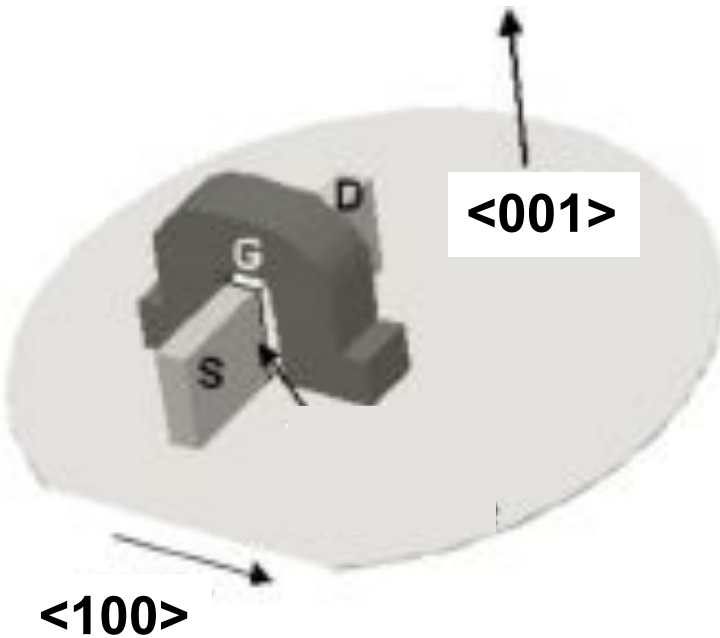
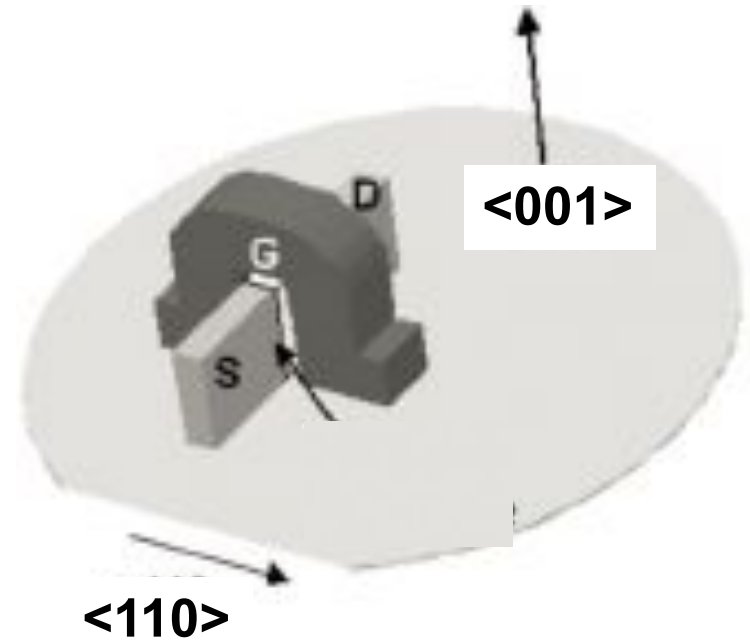
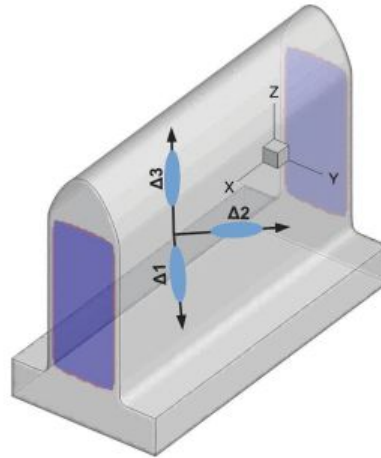
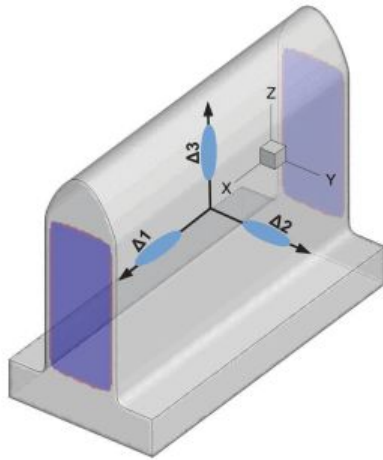
$$m_{\perp} = m_l = 0.916m_0, \quad m_{\parallel} = m_t = 0.196m_0$$

$\Delta_4$ -band:

$$m_{\perp} = m_t = 0.196m_0, \quad m_{\parallel} = (m_l m_t)^{1/2} = 0.42m_0$$



# FinFETs on rotated substrate



| Orientation           | Valley     | $1/m_{yy}^*$             | $1/m_{zz}^*$ | $m_{Tr}^*$      |
|-----------------------|------------|--------------------------|--------------|-----------------|
| $\langle 100 \rangle$ | $\Delta 1$ | $1/m_t$                  | $1/m_t$      | $m_l$           |
| $\langle 100 \rangle$ | $\Delta 2$ | $1/m_l$                  | $1/m_t$      | $m_t$           |
| $\langle 100 \rangle$ | $\Delta 3$ | $1/m_t$                  | $1/m_l$      | $m_t$           |
| $\langle 110 \rangle$ | $\Delta 1$ | $(m_t + m_l)/(2m_t m_l)$ | $1/m_t$      | $(m_t + m_l)/2$ |
| $\langle 110 \rangle$ | $\Delta 2$ | $(m_t + m_l)/(2m_t m_l)$ | $1/m_t$      | $(m_t + m_l)/2$ |
| $\langle 110 \rangle$ | $\Delta 3$ | $1/m_t$                  | $1/m_l$      | $m_t$           |

# Constant energy surfaces with different wafer orientation.

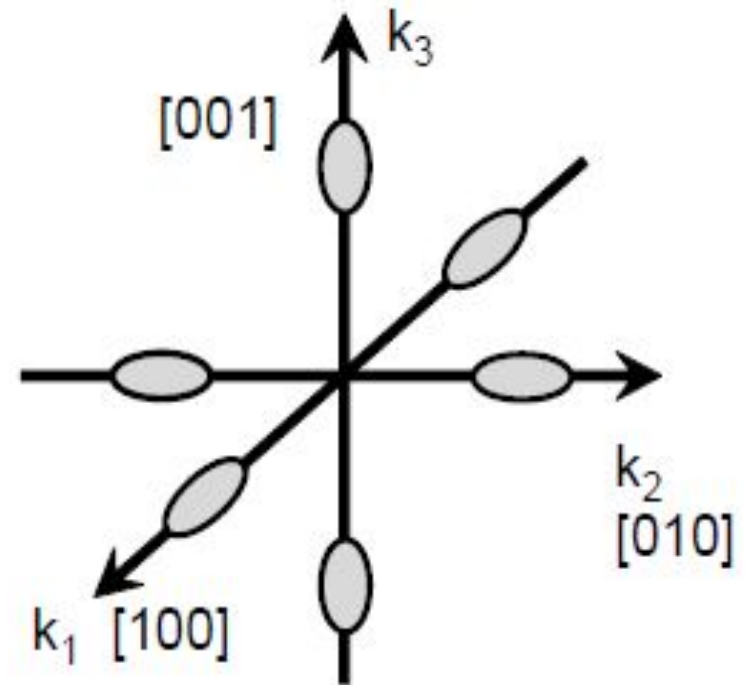
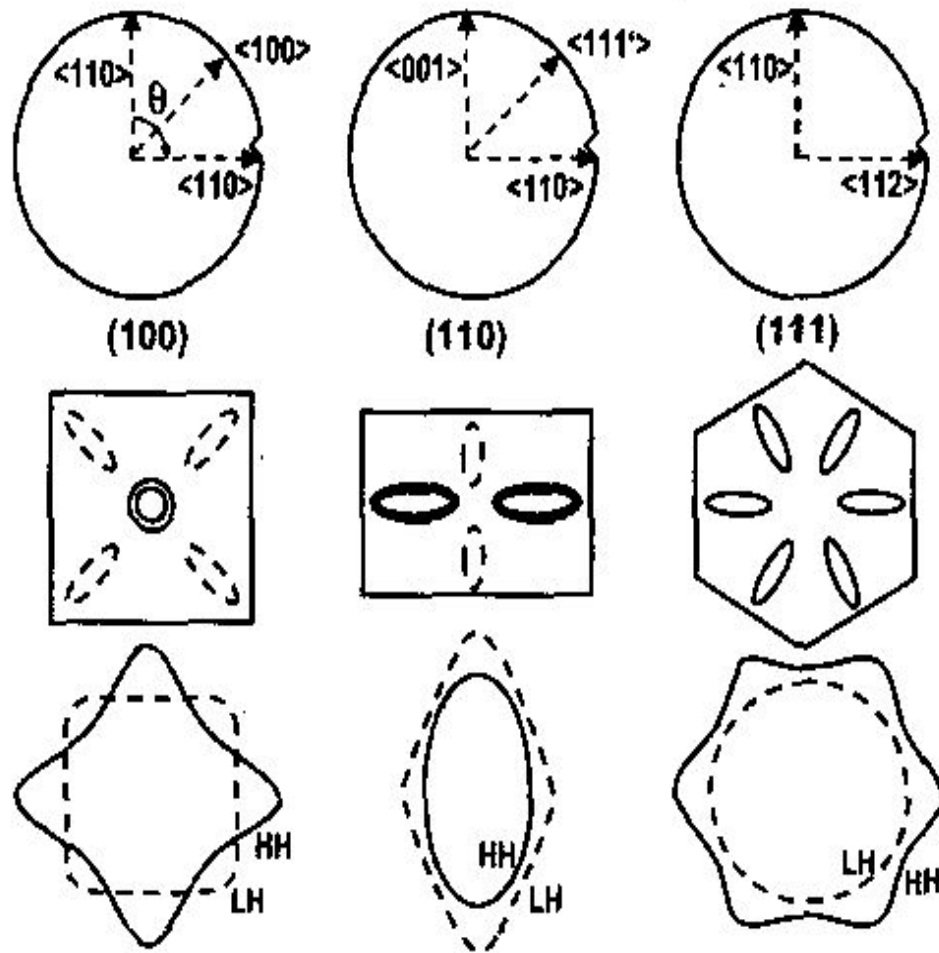
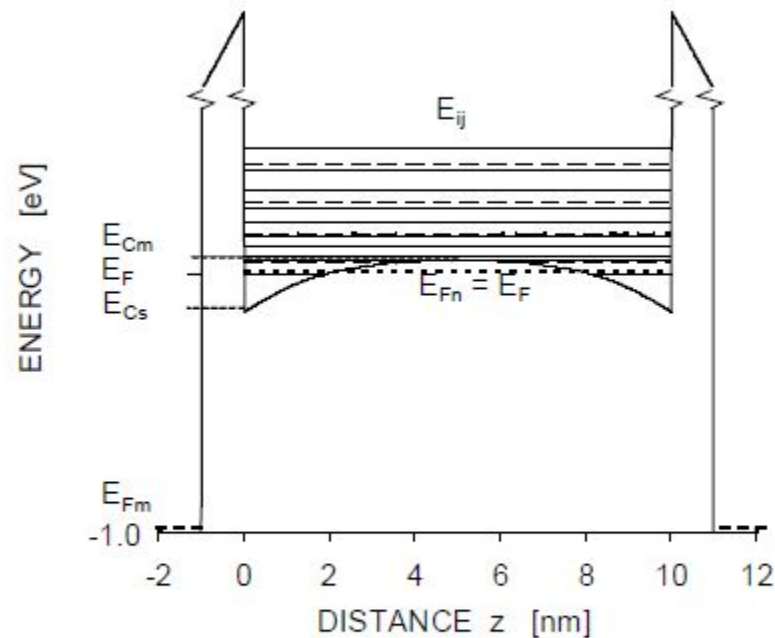


TABLE I. Transport, width, and confinement effective masses and subband degeneracies for three different technologically important semiconductor wafer orientations.

| (Wafer)/[transport]<br>/[width]       | Valley    | $m_x$                            | $m_y$          | $m_z$                | Degeneracy |
|---------------------------------------|-----------|----------------------------------|----------------|----------------------|------------|
| (001)/[100]/[010]                     | $\Delta$  | $m_t$                            | $m_l$          | $m_l$                | 2          |
|                                       |           | $m_l$                            | $m_t$          | $m_t$                | 2          |
|                                       |           | $m_l$                            | $m_l$          |                      | 2          |
| (111)/[ $\bar{2}11$ ]/[ $0\bar{1}1$ ] | $\Lambda$ | $m_l[(2m_l+m_t)/(m_l+2m_t)]$     | $(m_l+2m_t)/3$ | $3m_lm_t/(2m_l+m_t)$ | 4          |
|                                       | $\Delta$  | $(2m_l+m_t)/3$                   | $m_t$          | $3m_lm_t/(2m_l+m_t)$ | 2          |
|                                       |           | $(2/3)m_l[(2m_l+m_t)/(m_l+m_t)]$ | $(m_l+m_t)/2$  | $3m_lm_t/(2m_l+m_t)$ | 4          |
|                                       | $\Lambda$ | $m_t$                            | $m_t$          | $m_l$                | 1          |
|                                       |           | $(8m_l+m_t)/9$                   | $m_t$          | $9m_lm_t/(8m_l+m_t)$ | 1          |
|                                       |           | $(m_t/3)[(8m_l+m_t)/(2m_l+m_t)]$ | $(2m_l+m_t)/3$ |                      | 2          |
| (110)/[001]/[ $0\bar{1}0$ ]           | $\Delta$  | $m_t$                            | $(m_l+m_t)/2$  | $2m_lm_t/(m_l+m_t)$  | 4          |
|                                       |           | $m_l$                            | $m_t$          | $m_t$                | 2          |
|                                       | $\Lambda$ | $(m_l+2m_t)/3$                   | $m_t$          | $3m_lm_t/(m_l+2m_t)$ | 2          |
|                                       |           | $3m_lm_t/(2m_l+m_t)$             | $(2m_l+m_t)/3$ | $m_t$                | 2          |

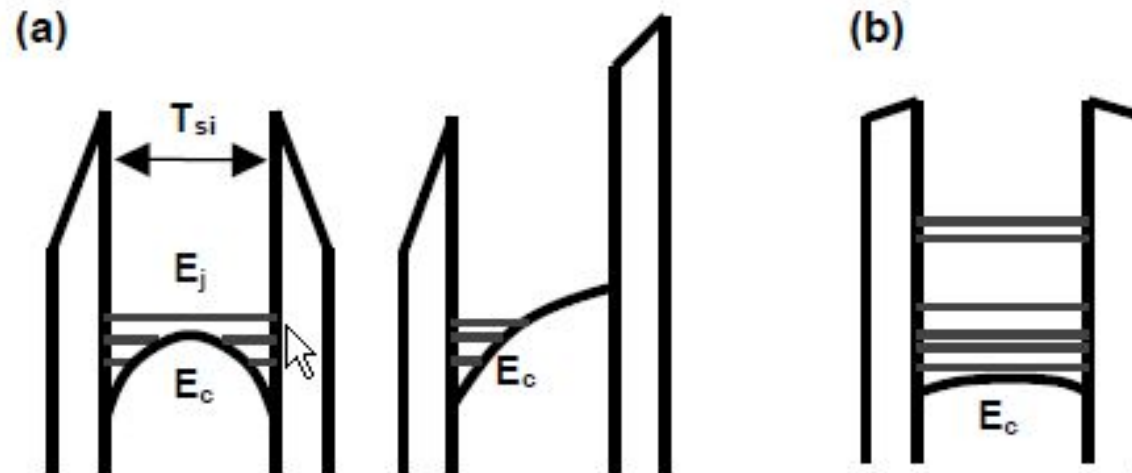
# Band diagram (FinFET)



**Fig. 4.7.** Band diagram of a symmetrical double-gate (DG) MOS system at  $V_G=1\text{V}$ .

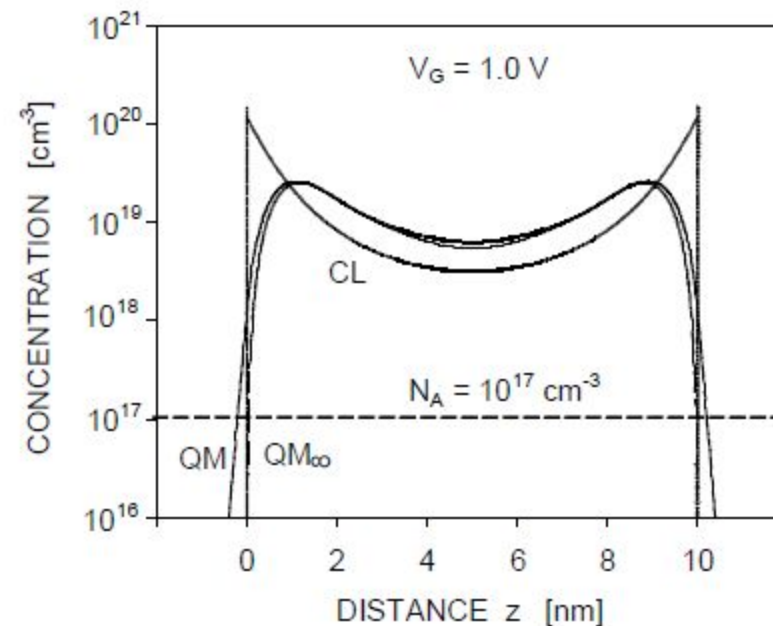
gate voltage  $V_G = 1\text{ V}$ . The (100)-oriented silicon layer has a thickness  $t_s = 10\text{ nm}$  and an acceptor atom concentration  $N_A = 10^{17}\text{ cm}^{-3}$ . The gate stack consist of a  $\text{SiO}_2$  layer having a thickness  $t_{ox} = 1\text{ nm}$  and a metal electrode with midgap work function  $\phi_M = 4.6\text{ eV}$ .

# Field vs structural confinement



**Fig. 3.10.** Energy-band diagrams of DG-MOSFET under different operation modes illustrate two mechanisms of carrier energy quantization in DG MOSFETs: (a) electrical confinement (EC); (b) structural confinement (SC).

# Classical vs quantum carrier distribution

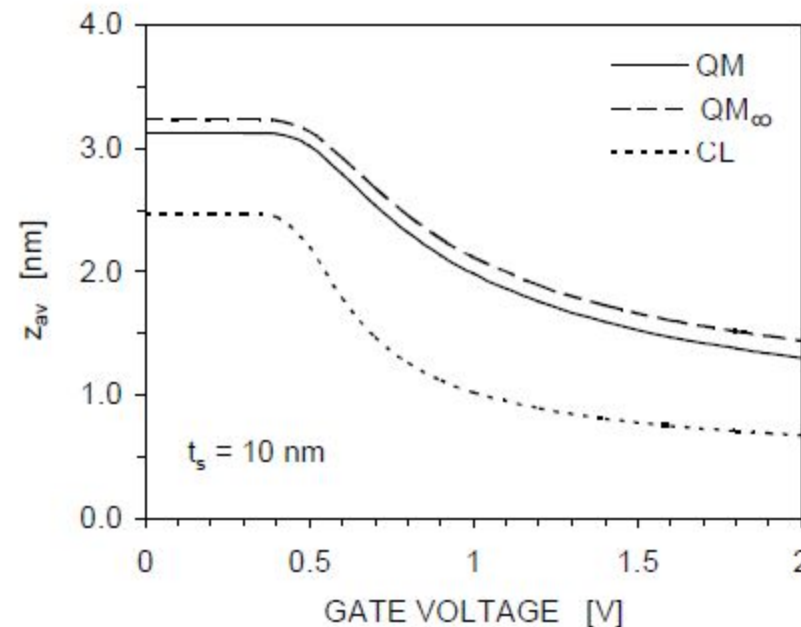


**Fig. 4.8.** Electron concentration distribution in the silicon body for different approaches: QM – quantum-mechanical, QM<sub>∞</sub> – quantum-mechanical with no penetration of the barrier region, CL – classical.

# Centroid of carriers

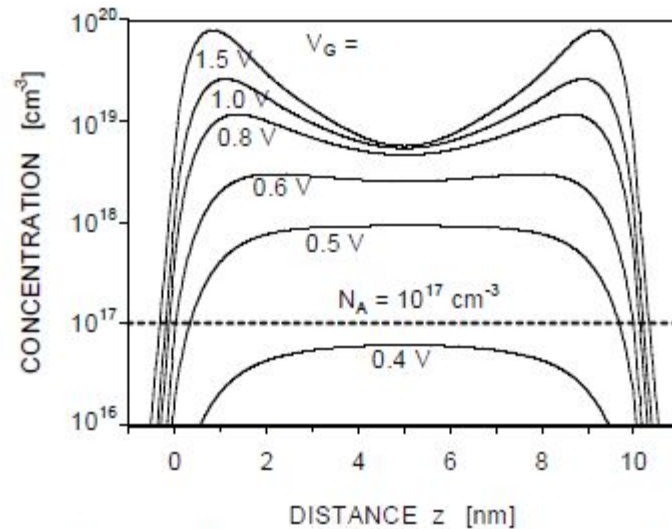
$$z_{av} = \frac{\int_0^{t_s/2} zn(z)dz}{\int_0^{t_s/2} n(z)dz} \quad (4.19)$$

As it can be seen in Fig. 4.10, the classical approach results in a significantly less average distance  $z_{av}$  due to the location of the peak of the electron concentration at the semiconductor surface.

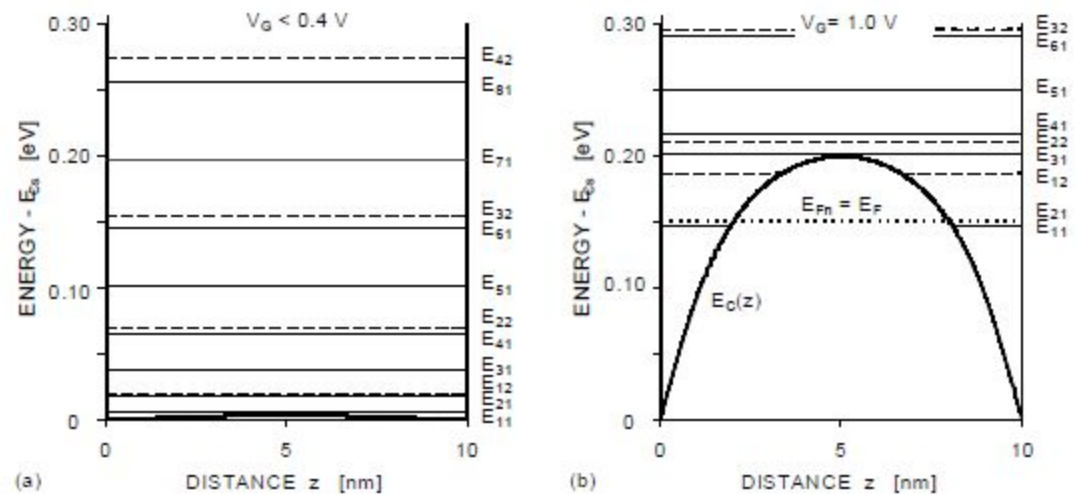


**Fig. 4.10.** Average distance of the electrons from the semiconductor surface for different modeling approaches.

# Bias dependence

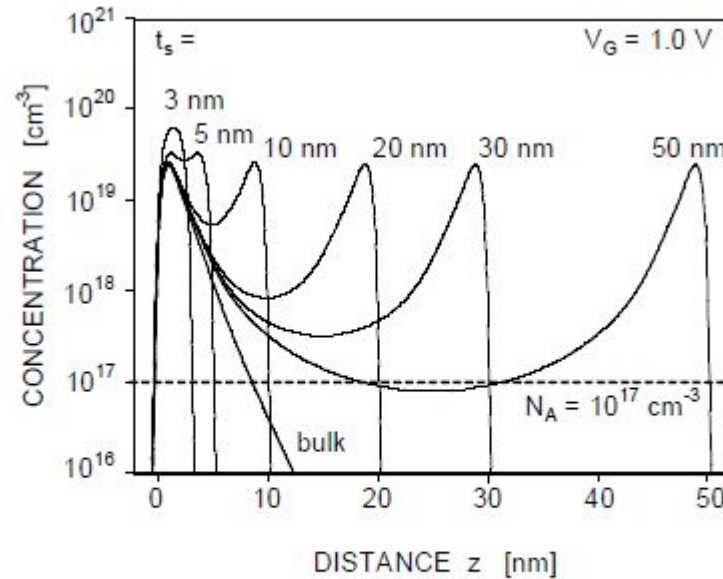


**Fig. 4.12.** Effect of gate voltage on electron concentration distribution in the silicon body of the symmetrical DG MOS structure



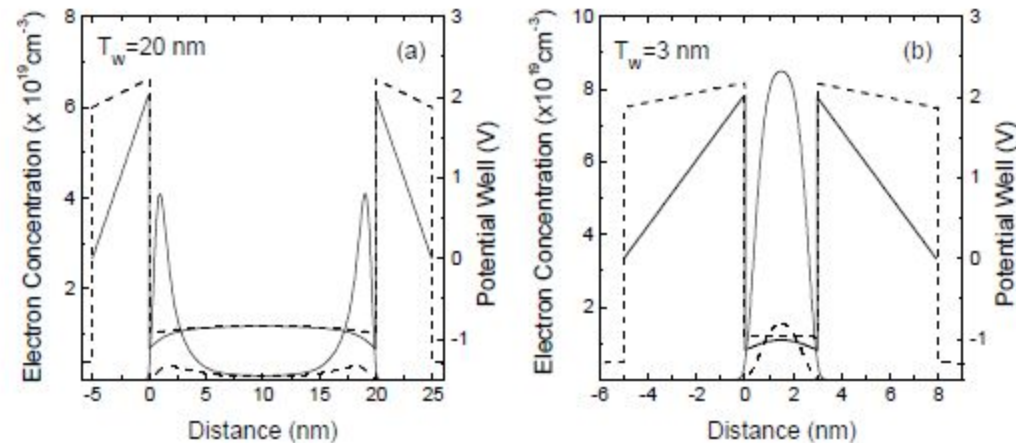
**Fig. 4.13.** Distribution of the conduction band edge and the electron energy levels  $E_{ij}$  at two gate voltages (a):  $V_G < 0.4\text{V}$ ; (b)  $V_G = 1\text{V}$ .

# $T_{si}$ dependence (Volume inversion)



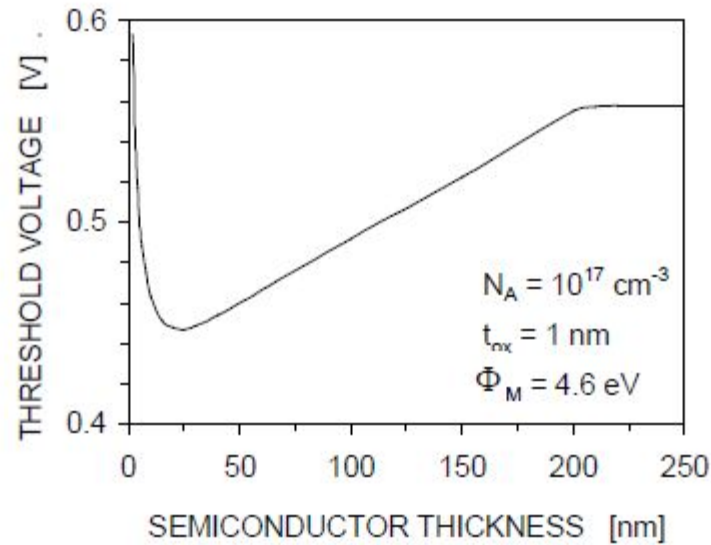
**Fig. 4.15.** Electron concentration profile at constant gate voltage ( $V_G=1$  V) for different values of semiconductor film thickness,  $t_s$ .

# $T_{si}$ dependence (Volume inversion)



**Fig. 5.2.** Electron distribution and potential well for two DGSOI devices with different silicon-layer thicknesses and two values of the inversion charge concentration. Dashed lines correspond to  $N_{inv} = 1 \times 10^{12} \text{ cm}^{-2}$ , and solid lines correspond to  $N_{inv} = 8 \times 10^{12} \text{ cm}^{-2}$ .

# Impact of $T_{Si}$ on $V_T$



**Fig. 4.20.** Threshold voltage of the doped DG MOS structure vs. semiconductor thickness.

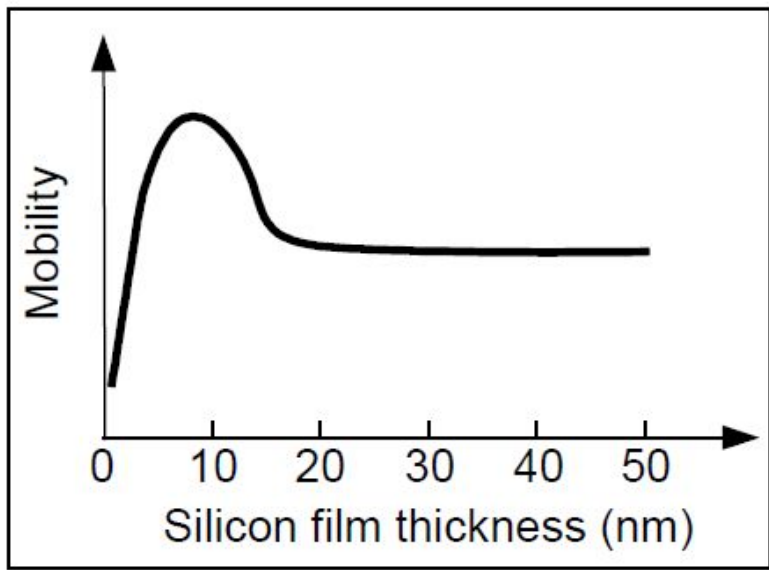


Fig. 1.20. Mobility vs. silicon film thickness in a double-gate transistor.

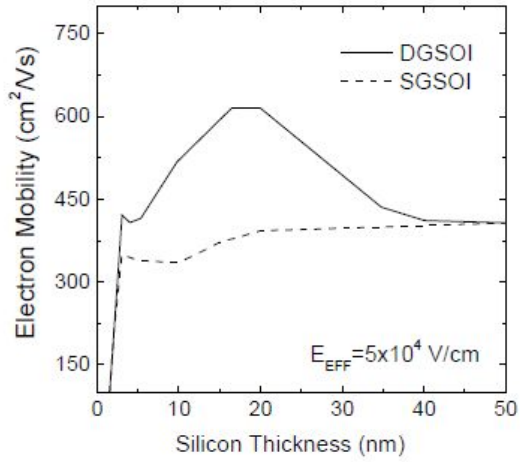
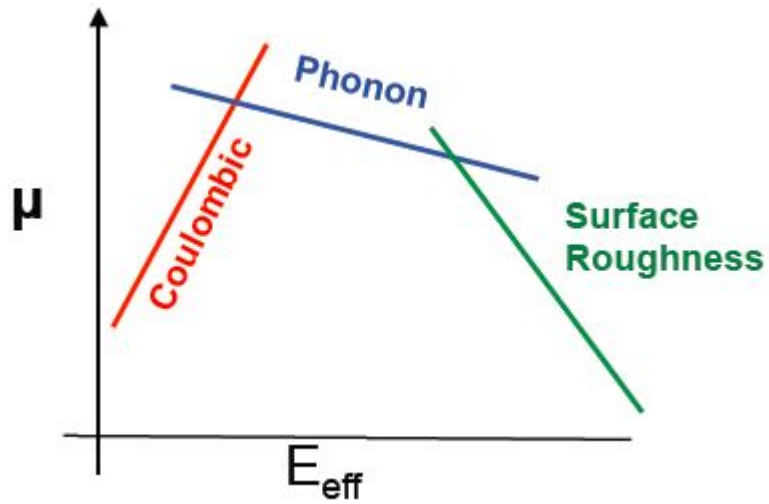
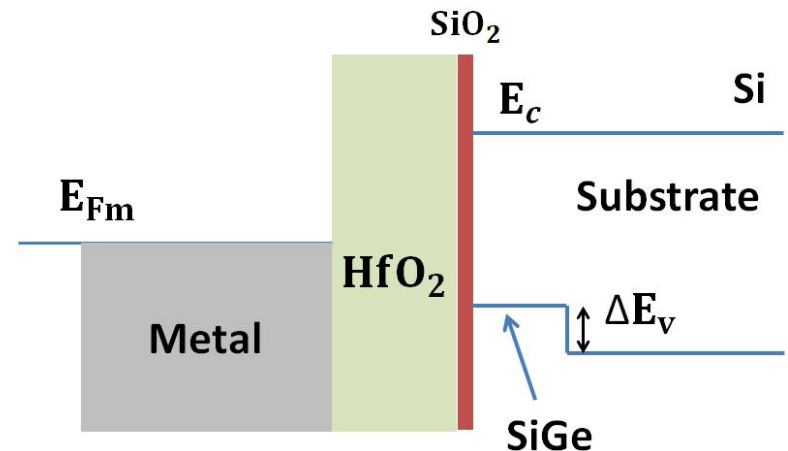
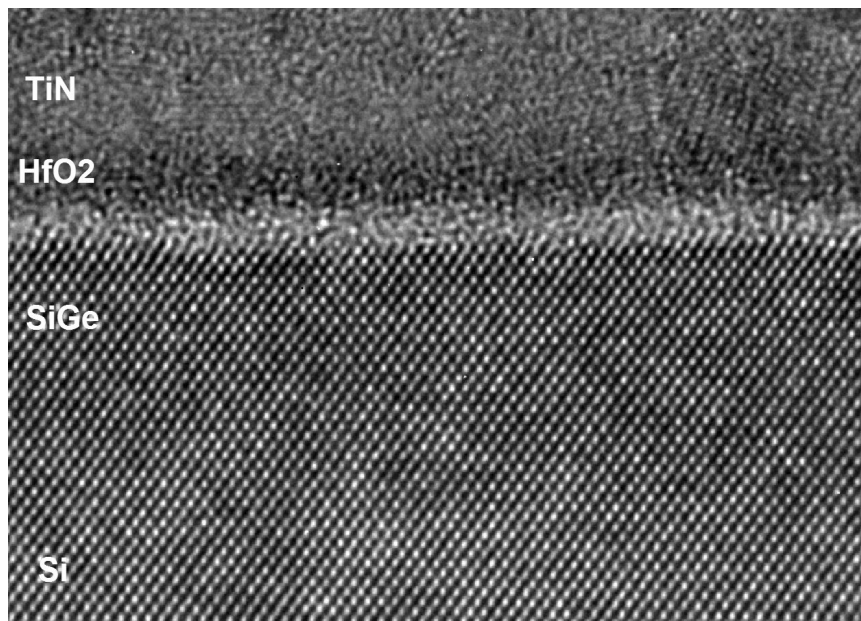


Fig. 5.14. Evolution of electron mobility for a DGSOI (solid line) and a SGSOI (dashed line) with the thickness of the silicon layer. All the scattering mechanisms were taken into account.

# **FinFETs with high mobility channel materials**

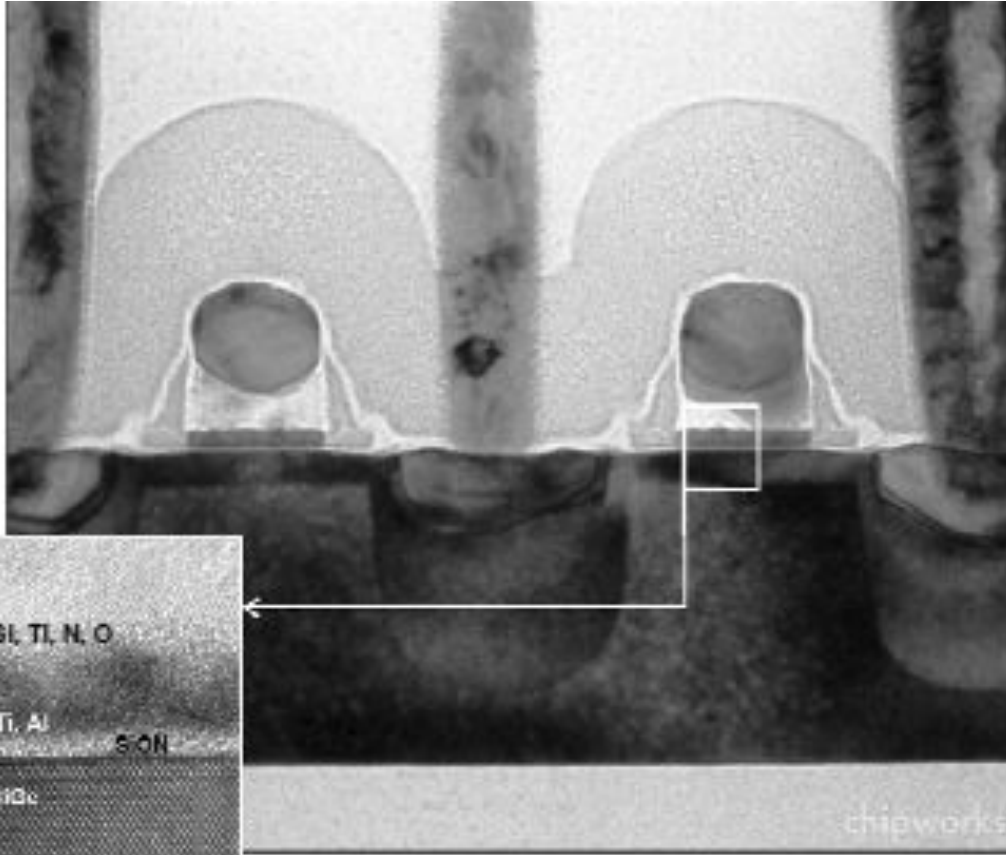
# Why SiGe channel?

- In gate first integration scheme, less choice for a PFET gate metal which can withstand S/D anneal
- With Silicon channel and a stable metal, the workfunction is away from valence band edge resulting in higher  $V_T$
- Solution- PMOS channel was changed from Si to SiGe
- Unexpected benefit – Better NBTI



# SiGe channel PFETs

- Gate first 32nm, 28nm bulk and 32, 22nm SOI technology

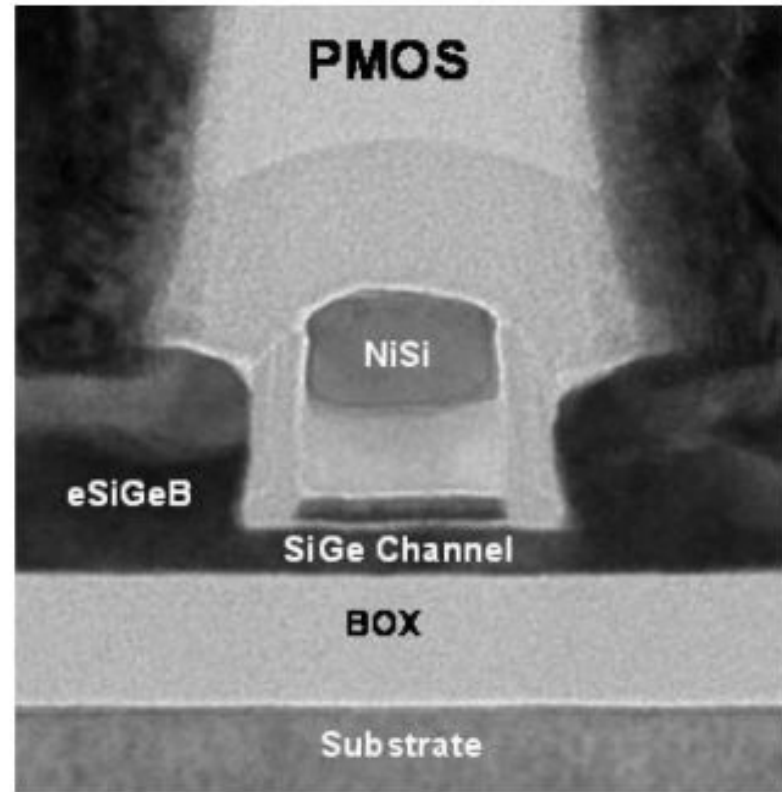
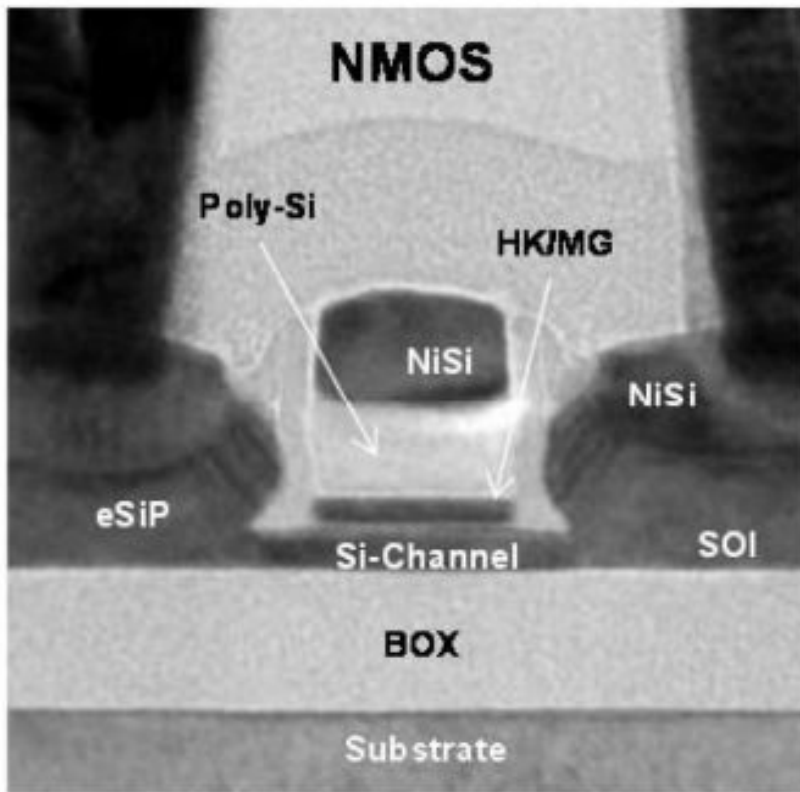


SiGe – The only non Silicon FET channel material which was used to manufacture VLSI chips in high volume

Apple A7 processor in iPhone 5S (Samsung 28LP)

# SiGe channel PFETs

- GlobalFoundries 22FDSOI technology



R. Carter et al, IEDM 2016

# No SiGe channel for next 3 nodes (20/14/10)

## 20nm RMG Bulk Planar FET

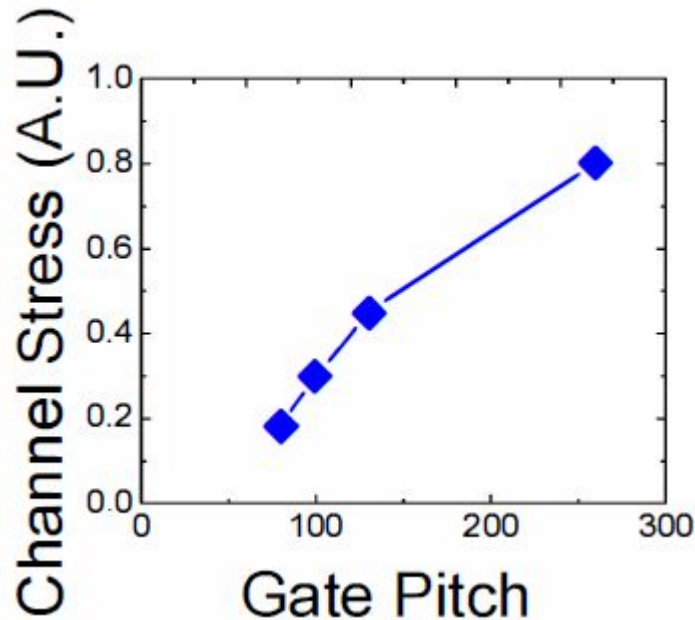
- Isolation (wells, Vt)
- IO EG gate growth
- Dummy gate patterning
- Logic SG, IO, SRAM extension/halo
- Spacer dep/patterning
- eSiGe formation
- Raised source/drain formation
- Advanced gate stack formation
- Tungsten stud (TS) contact formation
- CA/CB Tungsten contacts
- M1 double patterning
- Self Aligned V0
- BEOL

## 10nm RMG Bulk/SOI FinFET

- Active (Fin) Formation
- Dummy Gate Deposition & RIE
- Spacer Deposition & RIE
- eSiGe(P) /eSD (N) Growth
- ILD & CMP for Node Separation
- Dummy Gate Removal
- Multi-WF Gate Stack Formation
- Self Aligned Contact (SAC) Formation
- BEOL (Cu metallization)

# Reasons for revival of SiGe channel (1)

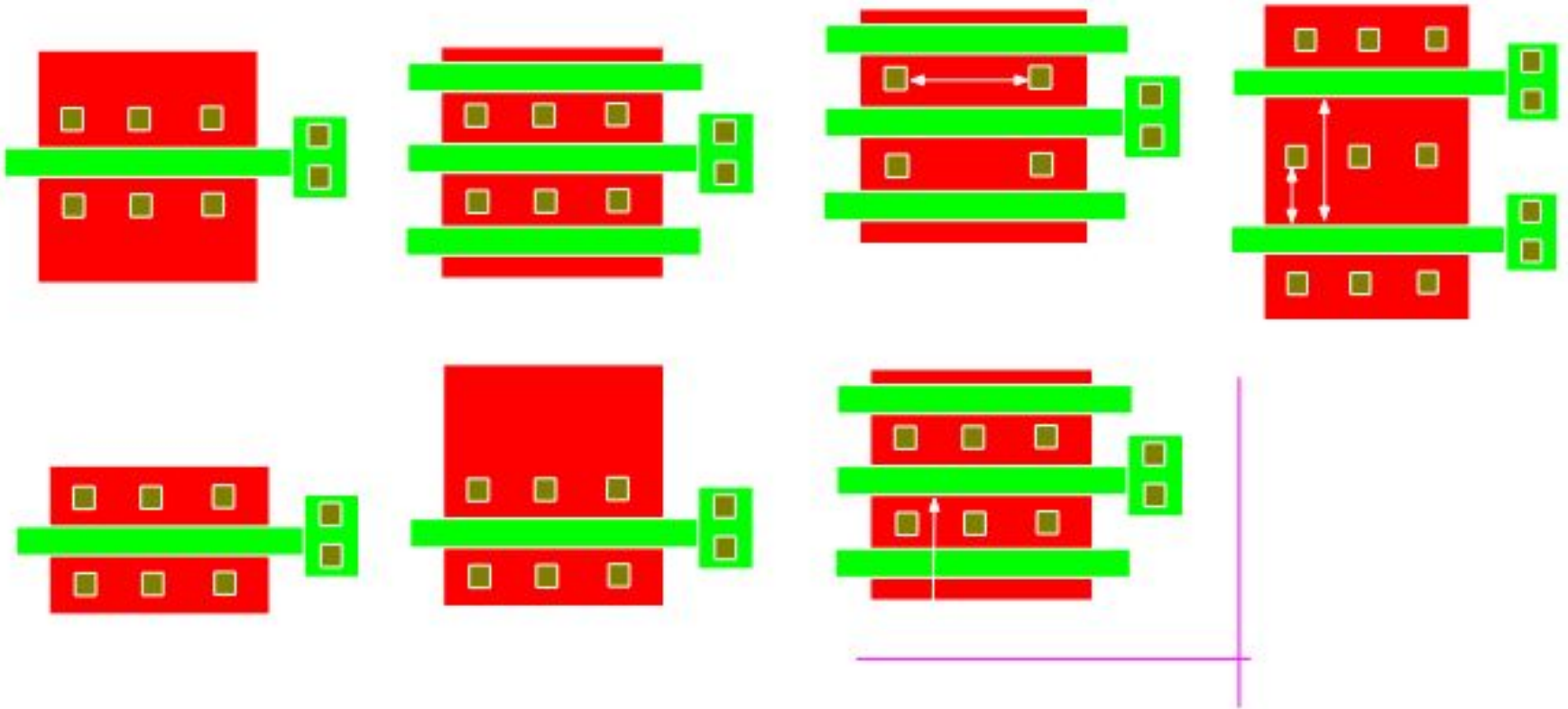
S/D stressor loses its effectiveness beyond the 14nm node



- Efficiency decreases with generation due to pitch scaling
- Need new techniques that increase stress in film (improve gap fill, recess, better eSiGe profile, increasing Ge %)

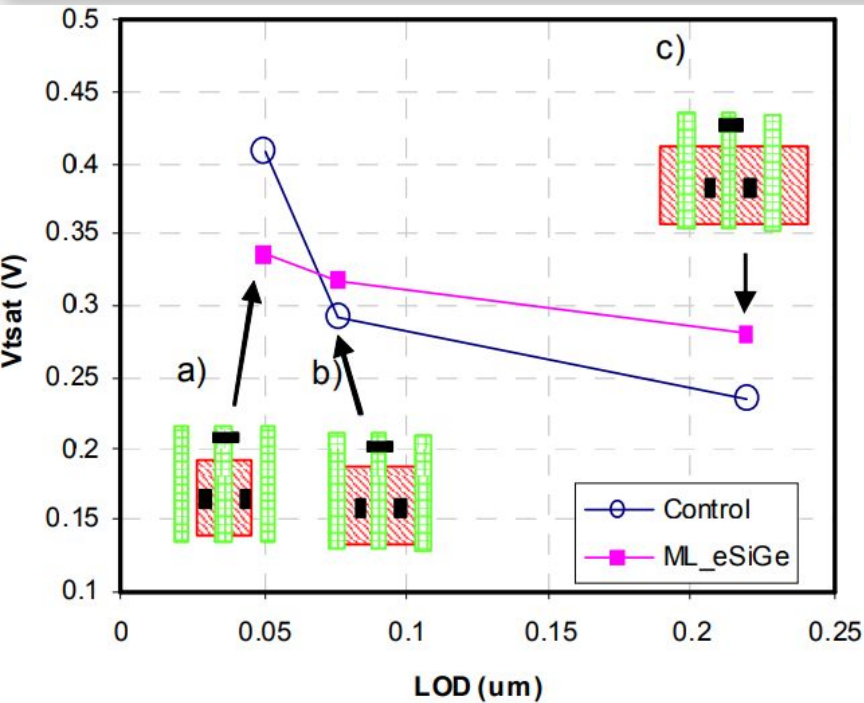
B. Doris et al, VLSI, 2013

# Reasons for revival of SiGe channel (2)



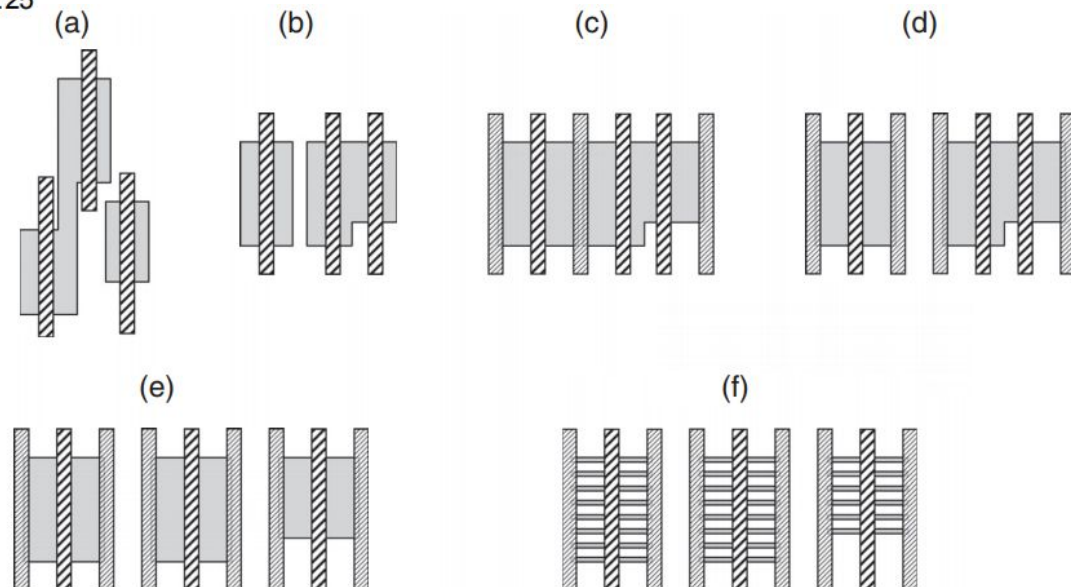
- Layout dependency - Stress introduced by CESL films or epi S/D is proportional to volume and its proximity to channel

# Reasons for revival of SiGe channel (2)



F. Sato et al, VLSI, 2013

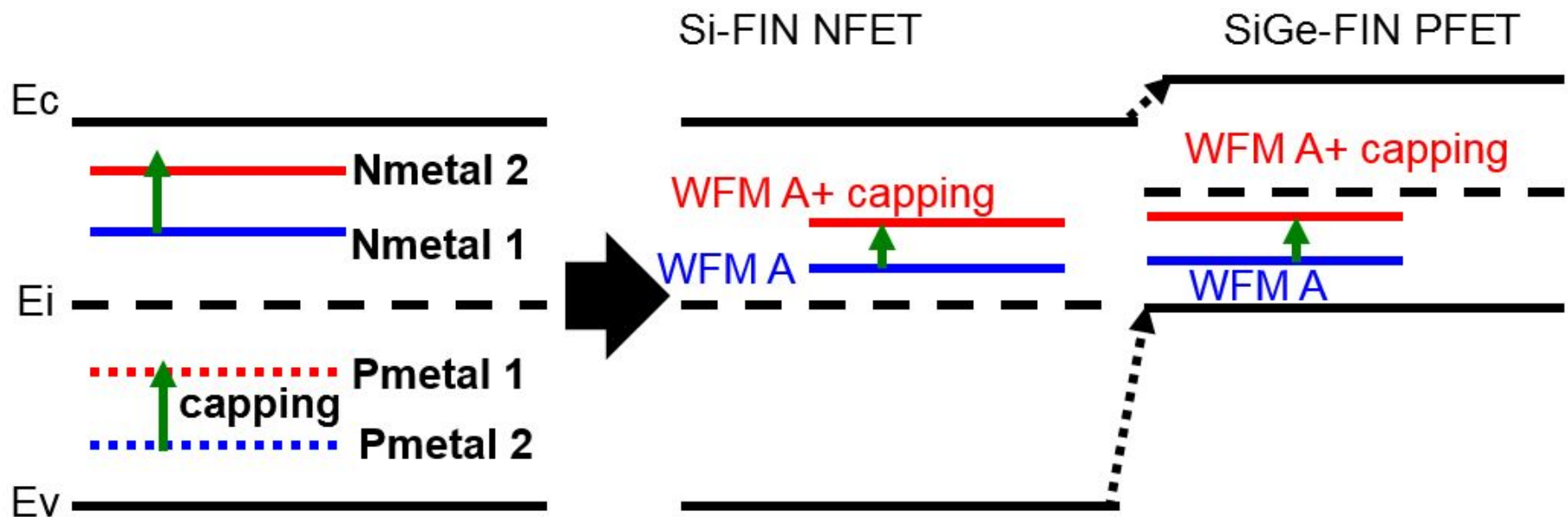
- Layout dependency – increases model complexity since 65nm
- Adds new layout restriction w/ every generation



L. Liebmann et al, CRC 2015

# Reasons for revival of SiGe channel (3)

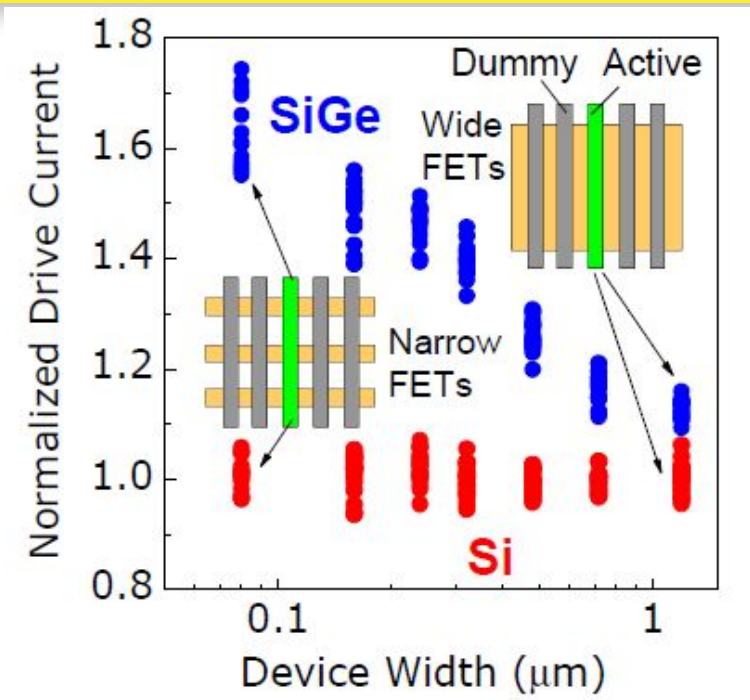
- Multi  $V_T$  devices - 1<sup>st</sup> gen FinFETs – combo of  $L_g$  and doping
- Next gen used diff metals – Dep and etching in very narrow RMG canyon
- Less # of metal layers with cSiGe



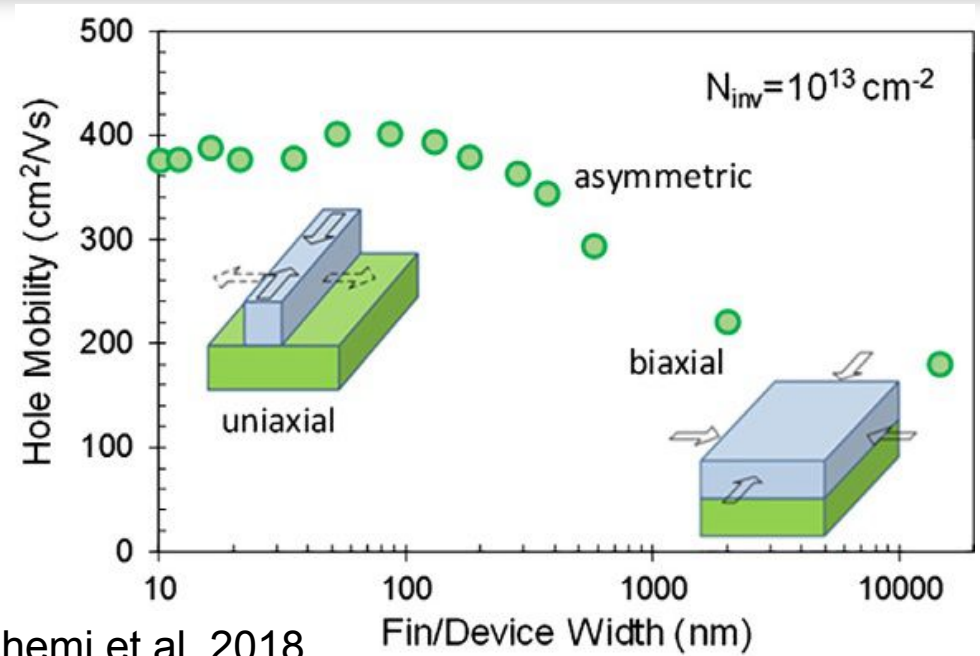
K. Seo et al, VLSI 2014

D. Guo et al, VLSI 2016

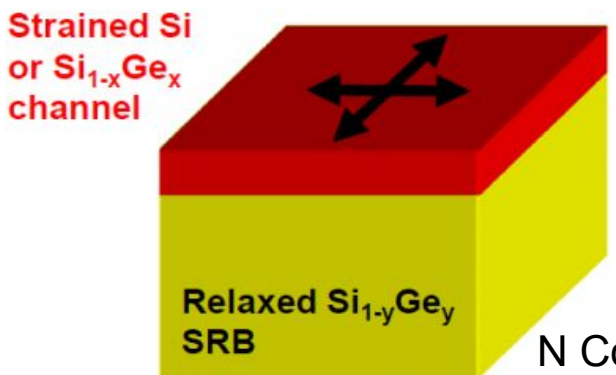
# Reasons for revival of SiGe channel (4)



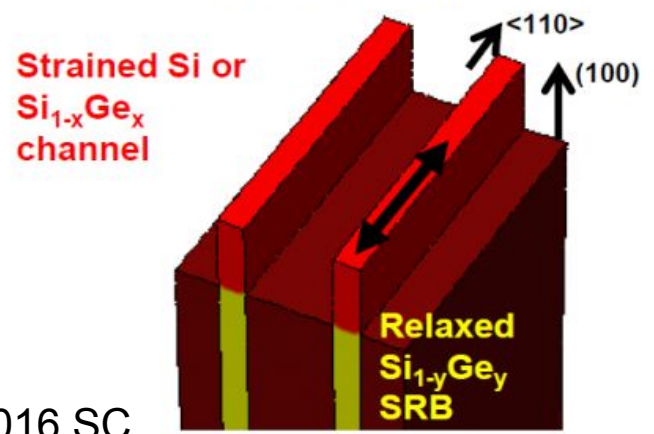
P Hashemi et al, 2018



K. Cheng et al, VLSI, 2012



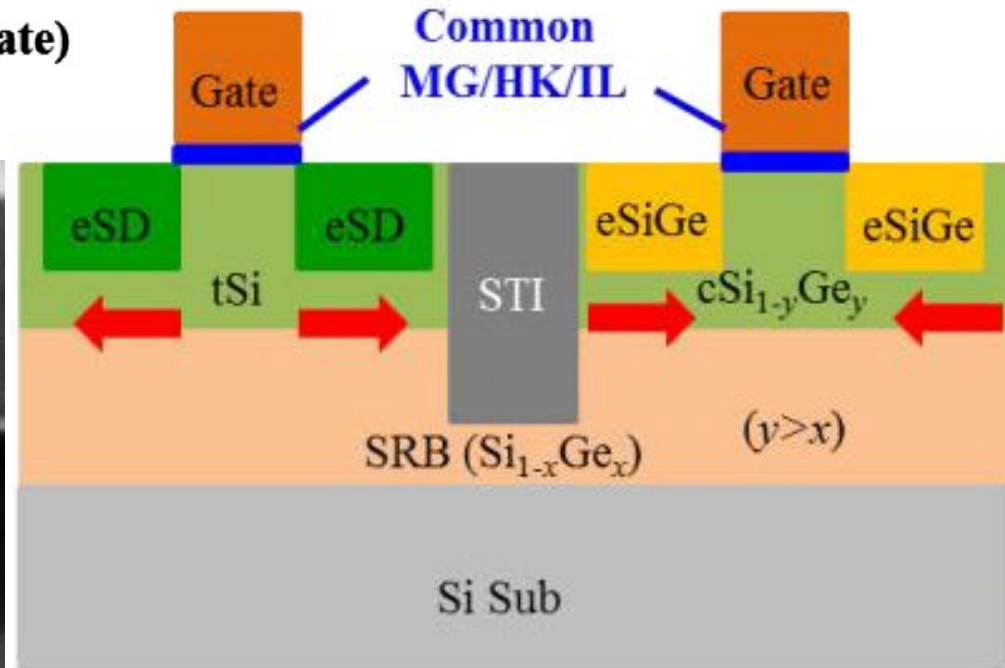
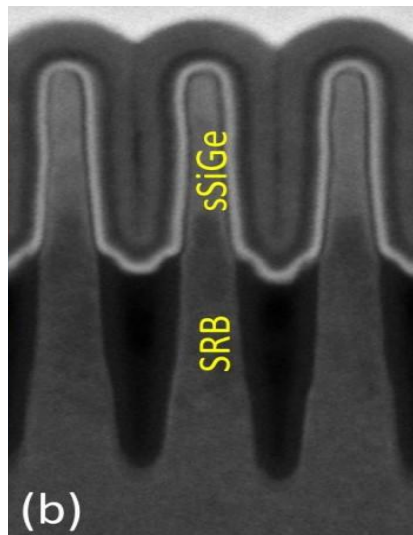
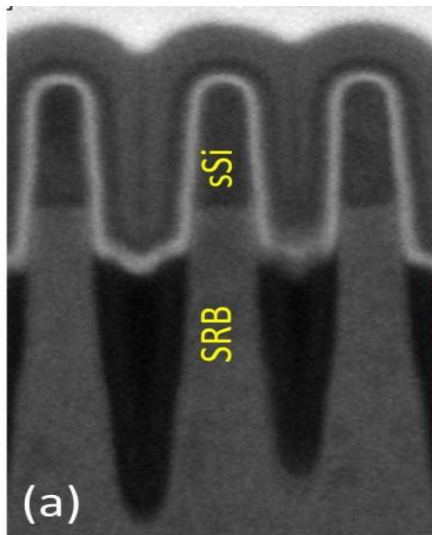
N Collaert, IEDM 2016 SC



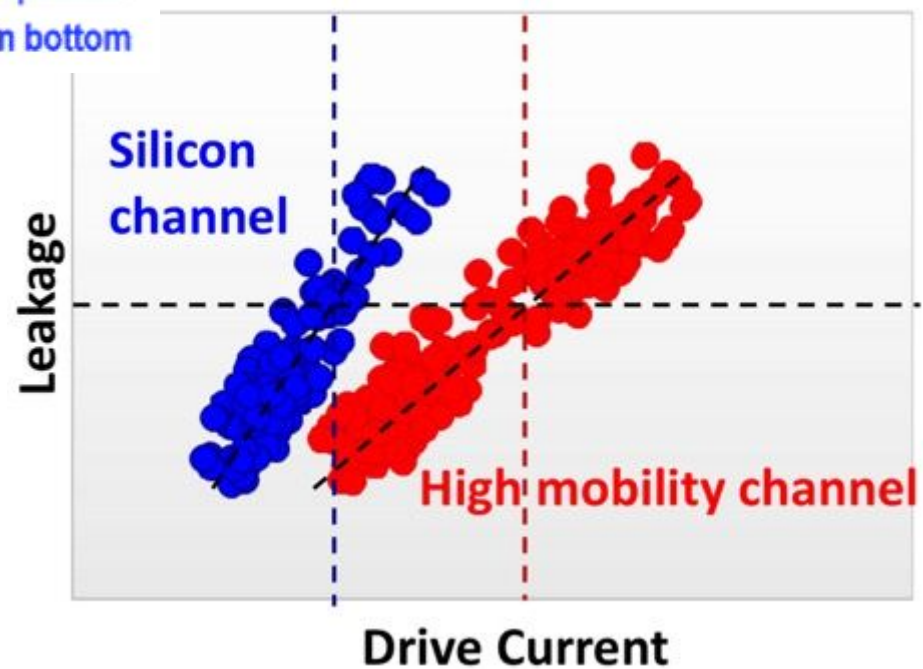
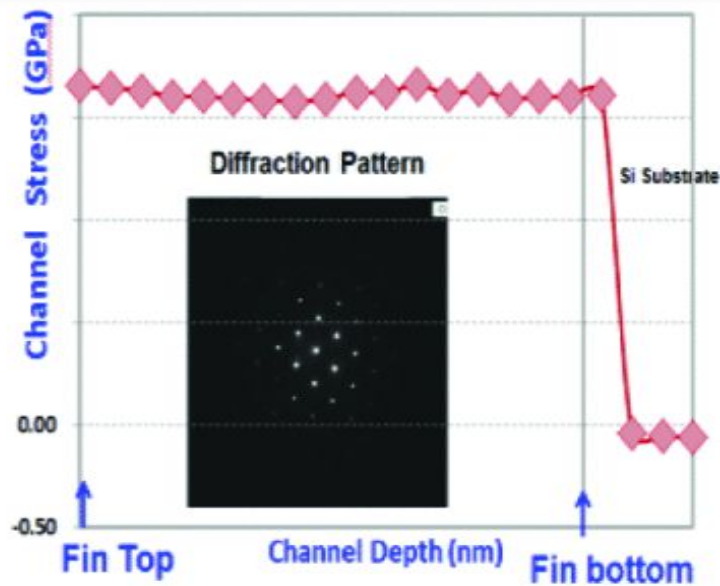
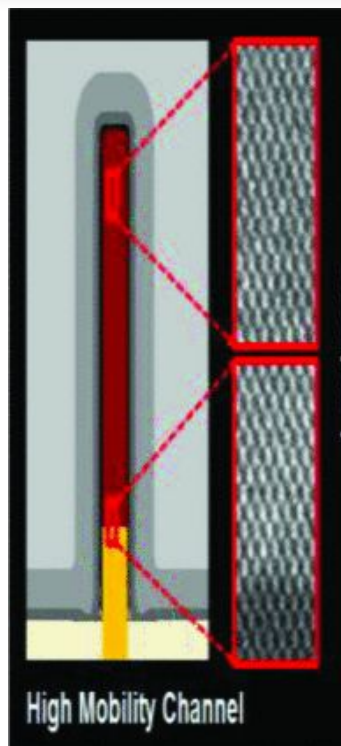
# SiGe FinFET process flow

- SiGe SRB / Si channel epitaxy
- Removal of Si channel in pFET region
- SiGe channel epitaxy in pFET region
- Novel Fin and STI
- S/D for stress (n/pFET)
- RMG (common HK/IL and metal gate)

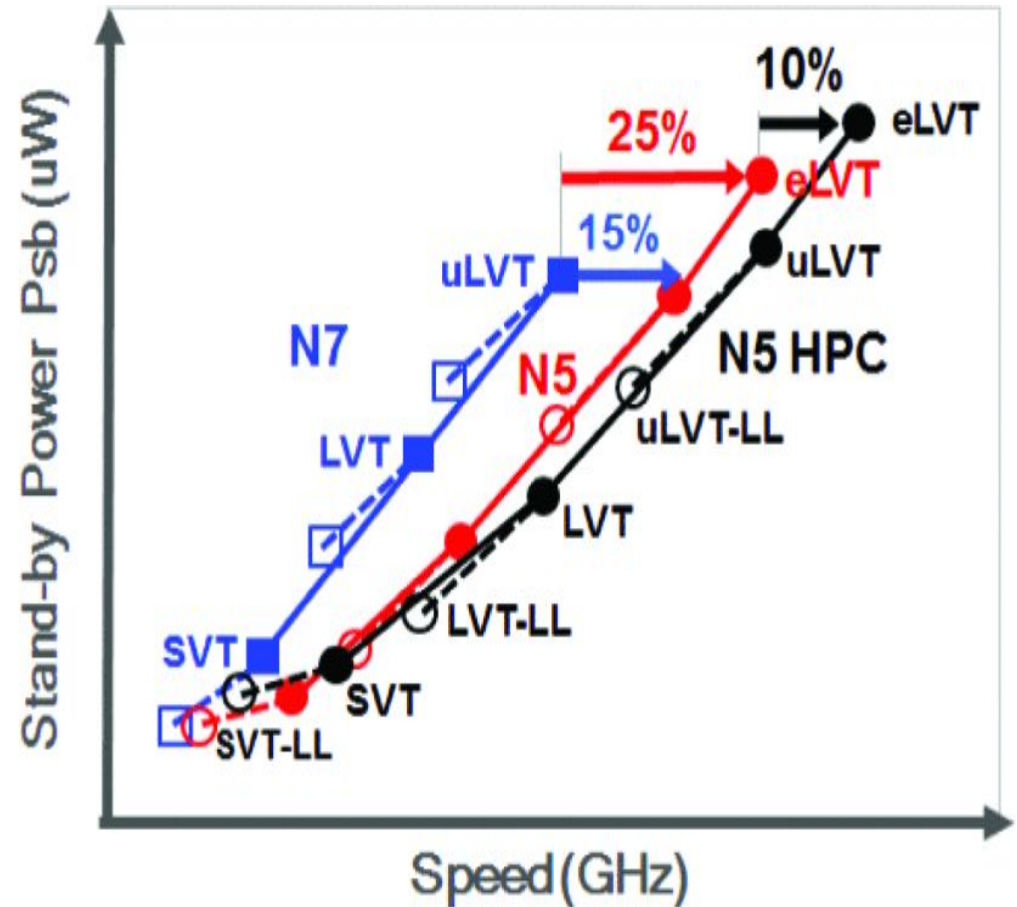
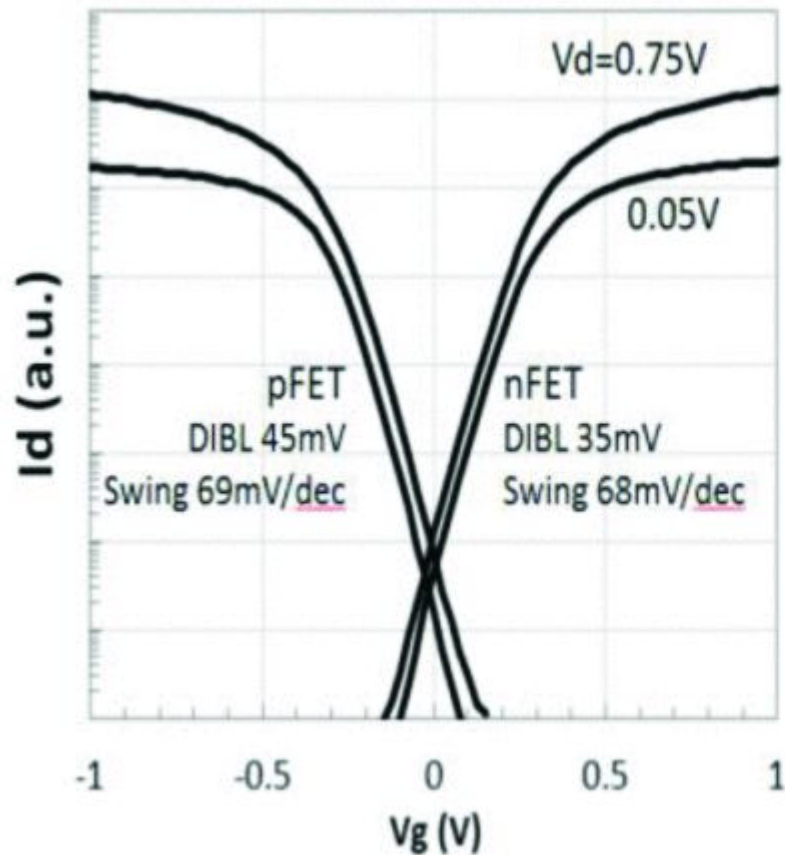
Requirement - Max strain in channel with low defectivity



# TSMC N5 PFET



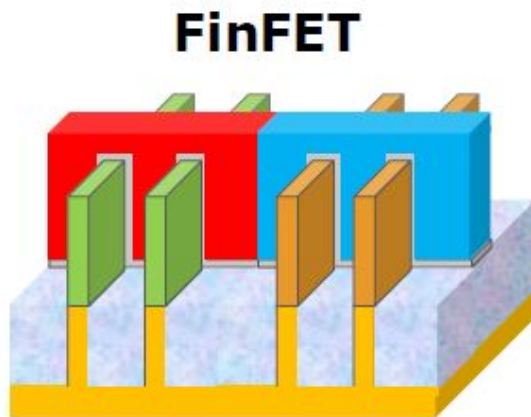
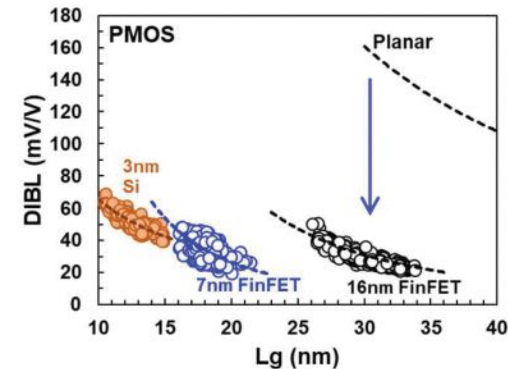
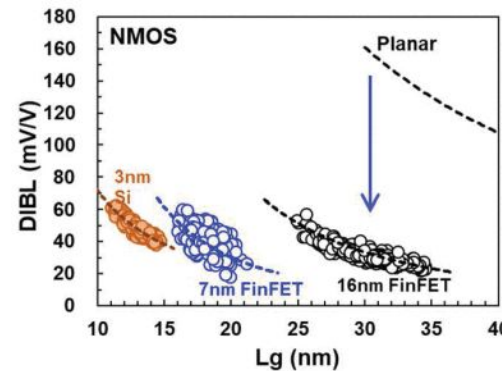
# TSMC N5



# What comes after FinFET?

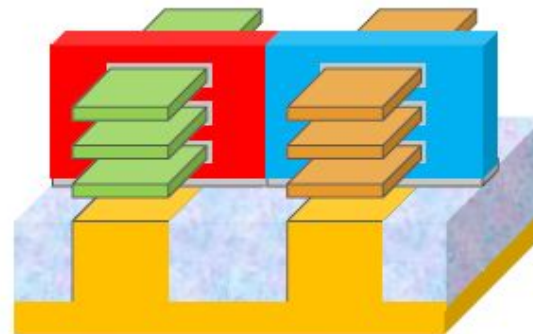
- FinFETs have been in production since 2011. TSMC 3nm node FET has a gate length  $> 10$  nm
- Needs nanosheets of Si with gate wrapped around on all sides for gate length scaling below 10 nm
- Nanosheets have also better current/footprint than nano wires. Nanosheet width ( $W_{\text{sheet}}$ ) is not quantized.

IEDM 2022



Auth *et al.*, Intel [14]

## Stacked Nanoribbons



Loubet *et al.*, IBM [24]



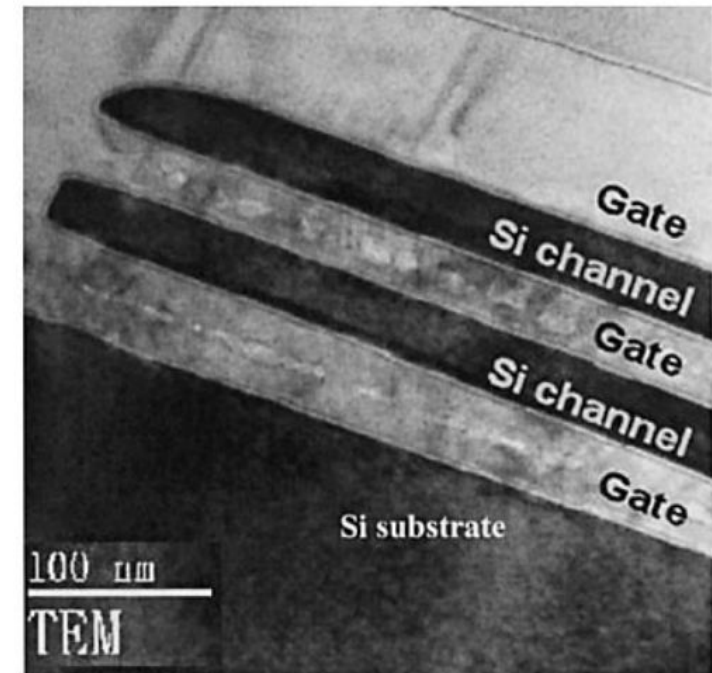
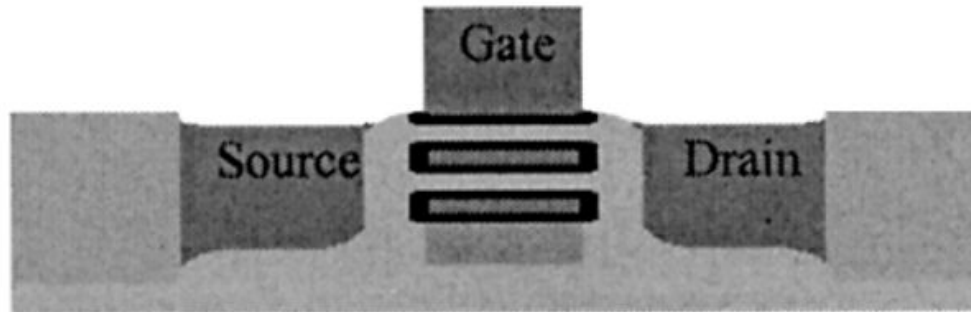
# 1st demo of nanosheet FET

IEEE TRANSACTIONS ON NANOTECHNOLOGY, VOL. 2, NO. 4, DECEMBER 2003

253

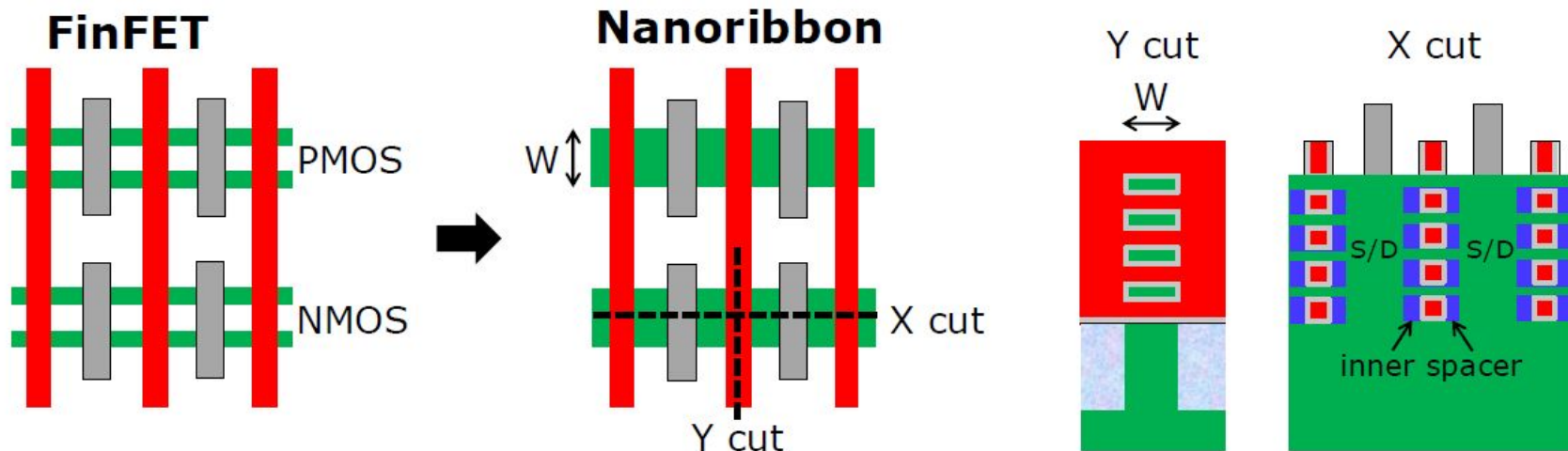
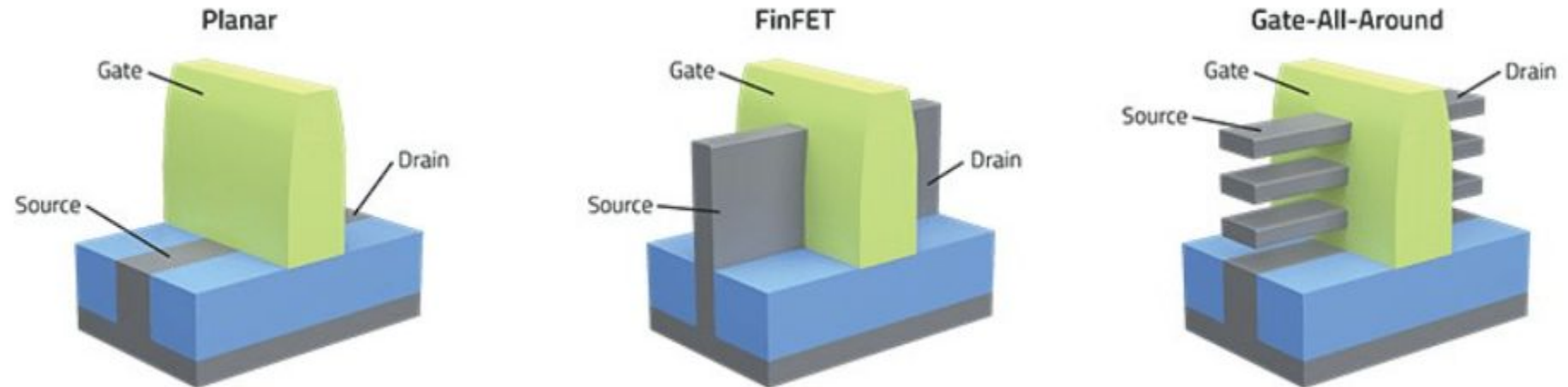
## A Novel Multibridge-Channel MOSFET (MBCFET): Fabrication Technologies and Characteristics

Sung-Young Lee, Sung-Min Kim, Eun-Jung Yoon, Chang-Woo Oh, Ilsub Chung, Donggun Park, and  
Kinam Kim, *Fellow, IEEE*



<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1264877>  
<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10529401>

# Different names - Nanosheet FET/ Ribbon FET/ MBC FET/ GAA FET



<https://newsroom.lamresearch.com/FinFETs-Give-Way-to-Gate-All-Around>

<https://spectrum.ieee.org/the-nanosheet-transistor-is-the-next-and-maybe-last-step-in-moores-law>

# Nanosheet transistor process flow (1)

- Si/SiGe epi (super-lattice)

- NS Fin patterning

- Isolation (STI)

- Fin/NS reveal (STI recess)

- Dummy Gate patterning

- S/D cavity & inner-spacers

- Junction S/D epi

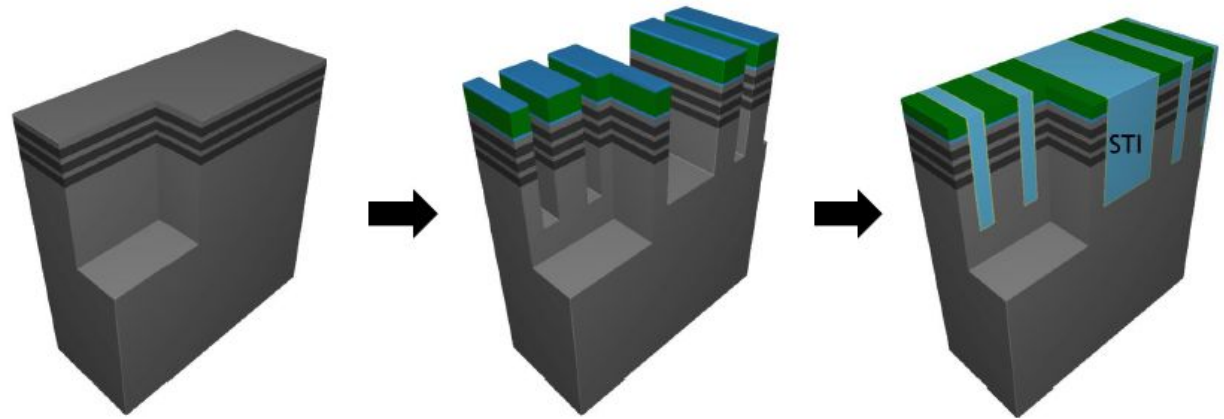
- ILD Ox & DG removal

- Nano-sheet release (SiGe etch)

- High-k Metal Gate

- Contact Engineering

- BEOL



## Si/SiGe Epitaxy Growth

- SiGe to be selectively etched away later

## Nanosheet Fin Patterning

- Fin width is variable

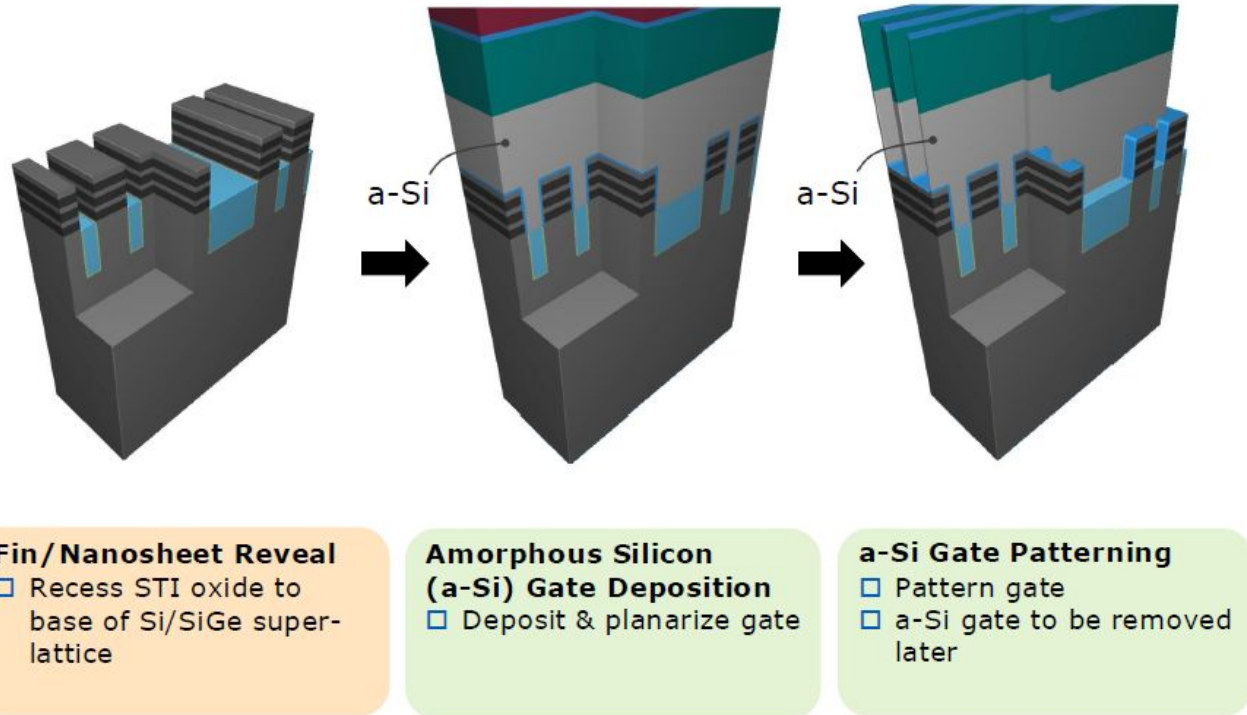
## STI Isolation

- STI dielectric gapfill
- STI oxide Chemical-Mechanical Polishing (CMP)

Yoshida, Applied Materials [25]

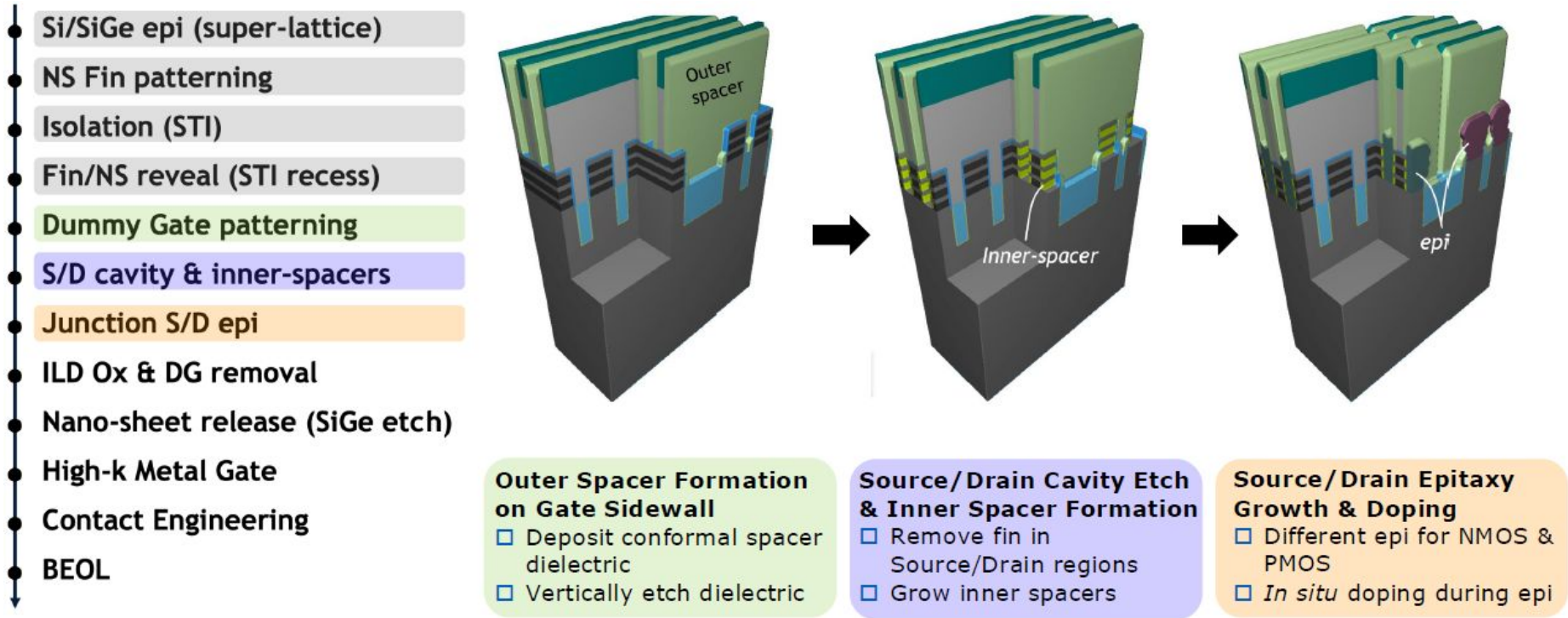
# Nanosheet transistor process flow (2)

- Si/SiGe epi (super-lattice)
- NS Fin patterning
- Isolation (STI)
- **Fin/NS reveal (STI recess)**
- **Dummy Gate patterning**
- S/D cavity & inner-spacers
- Junction S/D epi
- ILD Ox & DG removal
- Nano-sheet release (SiGe etch)
- High-k Metal Gate
- Contact Engineering
- BEOL



Yoshida, Applied Materials [25]

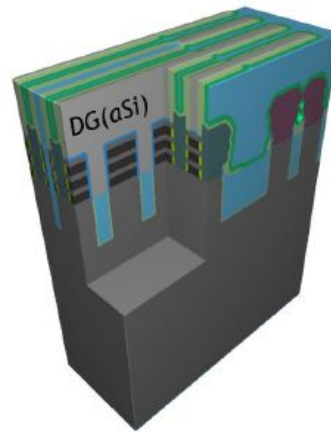
# Nanosheet transistor process flow (3)



Yoshida, Applied Materials [25]

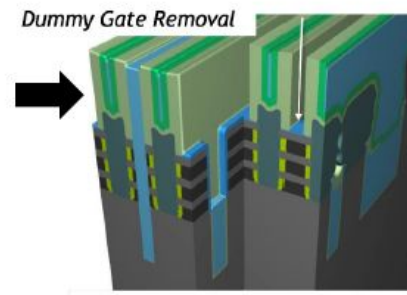
# Nanosheet transistor process flow (4)

- Si/SiGe epi (super-lattice)
- NS Fin patterning
- Isolation (STI)
- Fin/NS reveal (STI recess)
- Dummy Gate patterning
- S/D cavity & inner-spacers
- Junction S/D epi
- ILD Ox & DG removal
- Nano-sheet release (SiGe etch)
- High-k Metal Gate
- Contact Engineering
- BEOL



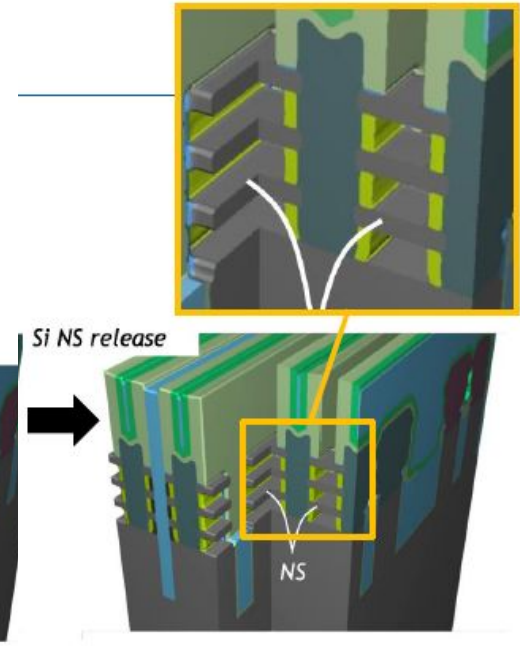
## ILD0 Oxide Deposition

- Deposit ILD0 contact dielectric
- Polish ILD0 to expose top of gate



## Dummy Gate Removal

- Etch away dummy amorphous silicon gate

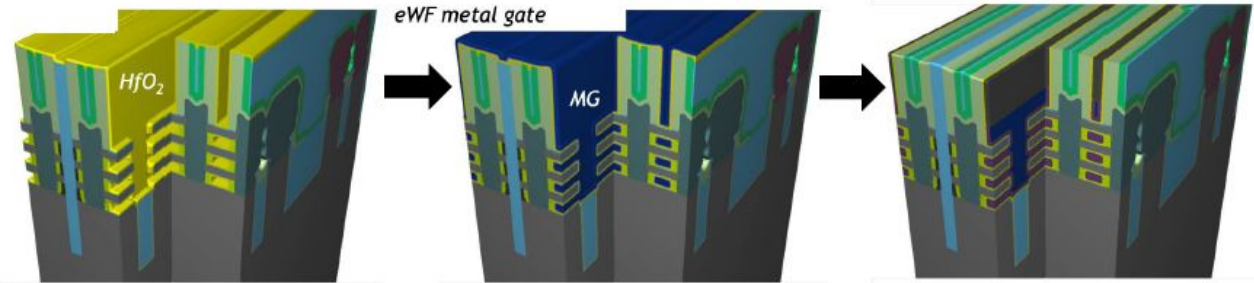


## Nanosheet Release

- Laterally etch exposed fin SiGe inside gate trench to suspend nanosheets

# Nanosheet transistor process flow (5)

- Si/SiGe epi (super-lattice)
- NS Fin patterning
- Isolation (STI)
- Fin/NS reveal (STI recess)
- Dummy Gate patterning
- S/D cavity & inner-spacers
- Junction S/D epi
- ILD Ox & DG removal
- Nano-sheet release (SiGe etch)
- **High-k Metal Gate**
- **Contact Engineering**
- **BEOL**



## High-K Deposition

- Conformally deposit high-K gate dielectric using ALD (Atomic Layer Deposition)

## Metal Gate Deposition

- Conformally deposit metal gate using ALD

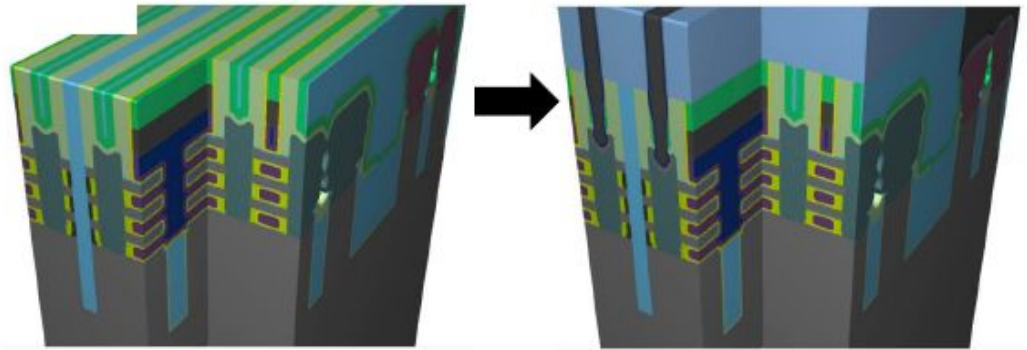
## Metal Fill Deposition & Gate Polish

- Conformally deposit metal fill using ALD
- CMP gate

Yoshida, Applied Materials [25]

# Nanosheet transistor process flow (6)

- Si/SiGe epi (super-lattice)
- NS Fin patterning
- Isolation (STI)
- Fin/NS reveal (STI recess)
- Dummy Gate patterning
- S/D cavity & inner-spacers
- Junction S/D epi
- ILD Ox & DG removal
- Nano-sheet release (SiGe etch)
- High-k Metal Gate
- **Contact Engineering**
- BEOL



## Gate Recess & Cap Formation

- Recess gate & backfill with dielectric for self-aligned diffusion contact

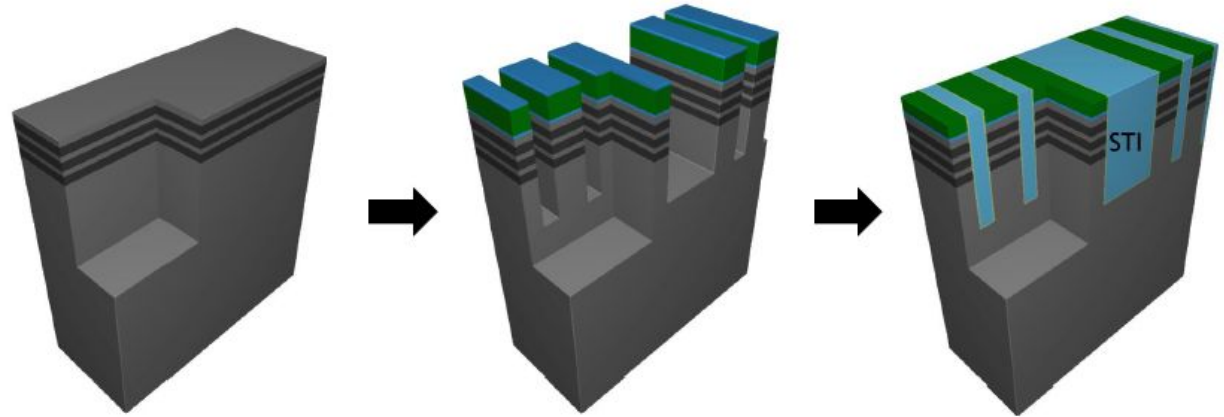
## Self-Aligned Gate Contact Etch, Fill & Polish

- Contact etch is insensitive to misalignment

Yoshida, Applied Materials [25]

# Nanosheet transistor process flow (7)

- Si/SiGe epi (super-lattice)
- NS Fin patterning
- Isolation (STI)
- Fin/NS reveal (STI recess)
- Dummy Gate patterning
- S/D cavity & inner-spacers
- Junction S/D epi
- ILD Ox & DG removal
- Nano-sheet release (SiGe etch)
- High-k Metal Gate
- Contact Engineering
- BEOL



## Si/SiGe Epitaxy Growth

- SiGe to be selectively etched away later

## Nanosheet Fin Patterning

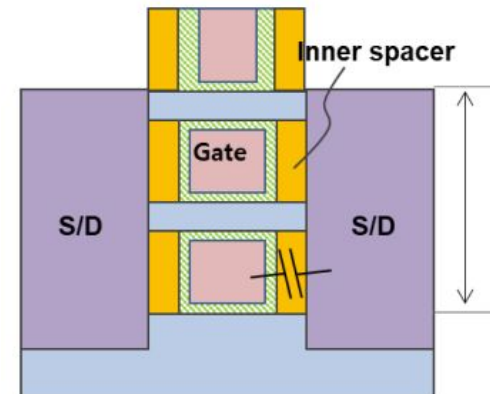
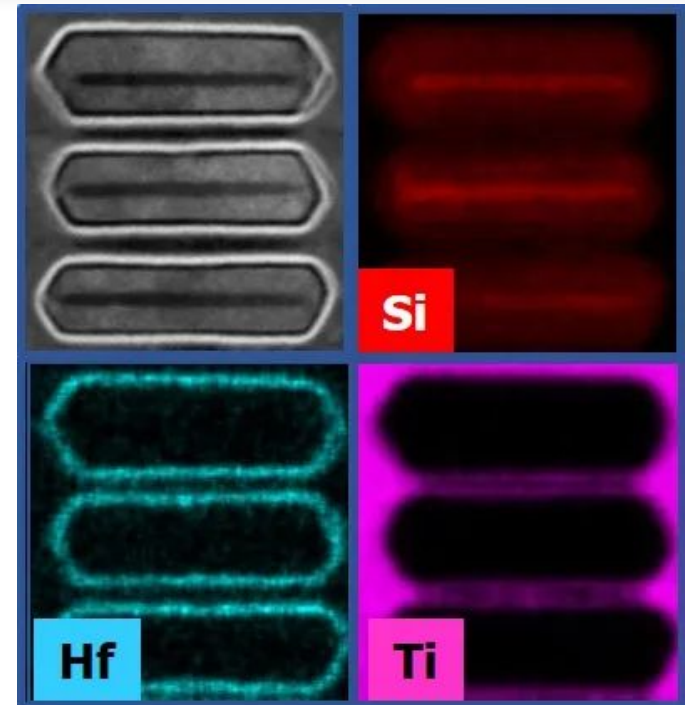
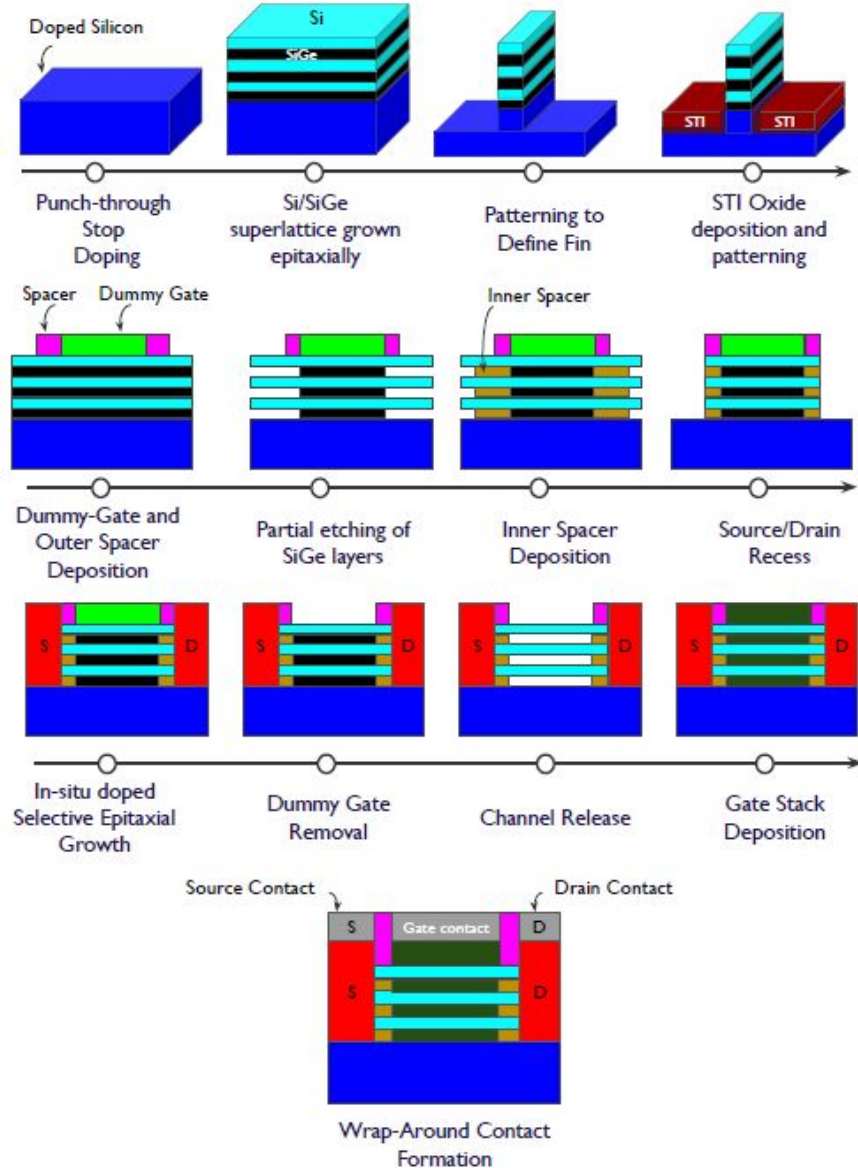
- Fin width is variable

## STI Isolation

- STI dielectric gapfill
- STI oxide Chemical-Mechanical Polishing (CMP)

Yoshida, Applied Materials [25]

# Nanosheet FET process flow



Inner spacer is placed between the inner gate and S/D to prevent shorting

# Future device roadmap (IMEC)

